

One Model, Many Geometries: Spectral Dynamics of Multilingual LLMs

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Abstract

The macro-level geometric co-evolution of languages during pretraining remains largely unstudied, despite its relevance to understanding cross-lingual capabilities. We track representation geometry via RankMe across 100+ pre-training checkpoints of two 8B multilingual LLMs and 14 diverse languages. We find that geometric phases are nearly identical across languages, yet a stable dimensionality hierarchy emerges early in pretraining and is robust across models with very different training scales. Tokenizer fertility partially explains this hierarchy, and early-training RankMe predicts both monolingual performance and cross-lingual transfer, making it an early and interpretable signal of multilingual capabilities.

1 Introduction

Multilingual training dynamics of LLMs have largely been studied using only feature-level methods (Blevins et al., 2022; Wang et al., 2024; Chang et al., 2022). However, Li et al. (2025) show that global spectral metrics can also be used to track the evolution of representation geometry during pretraining. We hypothesize that the macro-level co-evolution of languages predicts multilingual capabilities, and we explore this in two 8B multilingual LLMs using 100+ checkpoints and RankMe (Li et al., 2025; Garrido et al., 2023) as a measure of effective dimensionality.

We show that 14 diverse languages exhibit nearly identical geometric phases but differ in representational dimensionality. The cross-language RankMe hierarchy emerges within the first training checkpoints and persists across both models, suggesting it is largely architecture- rather than data-driven. We investigate whether differences stem from data or tokenizer factors, and we find a consistently negative correlation between fertility and RankMe. We also find evidence that early-training RankMe correlates with monolingual downstream performance,

and that RankMe differences between languages predict cross-lingual transfer abilities among them.

2 Approach

RankMe. Our primary metric for analyzing geometry evolution is RankMe, a measure of effective rank. For a layer L with hidden dimension D , we collect $N \gg D$ last-token activations from N distinct sequences into a matrix X . We then compute the centered covariance matrix Σ , and normalize its eigenvalues λ to yield a probability distribution $p_k = \lambda_k / \sum_j \lambda_j$. RankMe is the exponentiated entropy of p : $\exp(-\sum_k p_k \ln p_k)$.

Models. We use two open-source 8B multilingual LLMs: FuxiTranyu (Sun et al., 2024) (57 checkpoints) and Apertus (Apertus et al., 2025) (44 checkpoints).

Implementation Details. Since both models have $D = 4096$, we collect activations from $N = 10,000$ distinct sequences per language from FineWiki (Penedo, 2025), filtering and truncating them to have between 32 and 256 tokens. We select 14 languages to cover diverse families and representation in training data (see A.1), and we compute RankMe for each layer L and each checkpoint to track geometry evolution over pretraining.

3 Experiments

3.1 Layer-Depth Geometry

Beyond pretraining evolution, we study how representation geometry varies with *layer depth* and how this is across training and models.

Profiles, stability, and stratification. For each language and checkpoint, we compute RankMe (and PR, AlphaReQ, Top- K Variance) at every layer L , yielding a per-checkpoint *layer profile*. We test profile stability (H1) as the cosine similarity between each per-checkpoint profile and the checkpoint-mean profile, quantify cross-language

stratification (H2) as the standard deviation of RankMe across languages at each layer, and cluster languages (H3) via hierarchical clustering on mean layer-profile vectors.

Hierarchy origin and depth phases. To test whether the resulting hierarchy is architecture- or data-driven, we compute the per-layer Spearman ρ between Apertus and FuxiTranyu language rankings, and compare rankings from the first vs. last 5 checkpoints. Finally, we segment each model into depth phases at the inflection points of the combined normalized profile of all four metrics, and use the non-RankMe metrics to identify the spectral driver of each phase (Appendices B.4 and B.6).

3.2 Data Metrics as Geometry Predictors

We explore whether RankMe differences between languages can be explained by the training data share and tokenization-related metrics.

Tokenizer metrics. We compute **tokenizer fertility** and **vocabulary coverage**, two complementary tokenizer-level signals for each language from $N=5,000$ sampled Wikipedia. Fertility measures the average subword tokens per word (or per character for script-continuous Chinese and Japanese), and is computed as $\text{fertility}(\ell) = \frac{\sum_t |\text{tokenize}(t)|}{\sum_t |\text{units}(t)|}$. Vocabulary coverage measures the encoding *breadth*, and is computed as $\text{cov}(\ell) = |\mathcal{V}_\ell|/|\mathcal{V}|$. The two metrics are not significantly correlated (see Figure 22), confirming that they can capture orthogonal aspects of tokenization efficiency.

Correlation with training distribution. For FuxiTranyu, the full training language distribution is available, allowing us to study the correlation between RankMe values, exploring whether the amount of training data directly correlates with the quality of learned representations per-language.

Correlation analysis. For each layer at each training checkpoint, we compute Pearson r and Spearman ρ between each tokenizer metric and RankMe across the selected languages to identify correlations and analyze whether they evolve predictably during training or across layers.

3.3 RankMe and Downstream Performance

We explore whether RankMe correlates with downstream monolingual performance, evaluating four benchmarks at every training checkpoint: **m-MMLU** (Hendrycks et al., 2020; Lai et al.,

2023) (5-shot, 8 languages), **XCOPA** (Ponti et al., 2020) (0-shot, 5 languages), **Belebele** (Bandarkar et al., 2024) (5-shot, 10 languages), and **M-ARC** (Alexandra Institute, 2025) (5-shot, 9 languages).

Predictor and correlation analyses. From each language’s RankMe trajectory at the layer selected per model (layer 21 for FuxiTranyu, layer 19 for Apertus; see Appendix E.1), we extract 7 scalar predictors: RankMe at the first checkpoint, at checkpoint 10, and at the last checkpoint; mean RankMe over the first 10 checkpoints; and means over Q2–Q4 training windows. These are correlated with two outcomes per task: (i) *grokking onset* in billions of tokens, and (ii) *peak accuracy*. Grokking onset is defined as the first checkpoint where task accuracy exceeds the random-chance baseline by >15 percentage points and remains there for at least 2 consecutive checkpoints. Complementarily, at each checkpoint we compute the Spearman correlation between RankMe and accuracy for all benchmark languages.

3.4 RankMe and Cross-Lingual Transfer

We investigate whether differences in languages’ RankMe correlate with *cross-lingual transfer*, i.e., the models’ ability to apply knowledge learned in one language to perform tasks in another one.

Task. We find that the benchmark ECLeTic (Goldman et al., 2025) produces noisy transfer scores due to its limited sample size (see D.1), so we design our own task by adapting XNLI (Conneau et al., 2018), a 3-class natural language inference task. We evaluate models via in-context learning: given k demonstrations in a source language s , the model classifies a premise–hypothesis pair in a target language t . As XNLI is a parallel corpus, we use identical context and test examples across all language pairs ($N = 1000$ per pair). We define the *transfer score* as $\tau(s, t) = \text{acc}(s, t)/\text{acc}(s, s)$, i.e., the target accuracy normalized by the monolingual baseline.

Correlation analysis. For each language pair (s, t) at checkpoint C , we derive a predictor as either the difference or ratio of RankMe values for s and t , which we average across transformer layers and checkpoints up to C . We then measure the Spearman correlation between this predictor and $\tau(s, t)$ evaluated T checkpoints later, for $T \in \{0, 3, 9, 15, 30\}$. We consider a total of 90 language pairs by excluding same-language pairs.

4 Results and Analysis

4.1 Geometric Evolution of Languages

Figure 1 shows the last-layer RankMe evolution of FuxiTranyu over all pretraining checkpoints. All 14 languages follow the same two-phases geometric evolution, differing only in their absolute effective dimensionality. We find almost identical results for Apertus (see Figure 6), and we provide a mathematical characterization of these results in A.3.

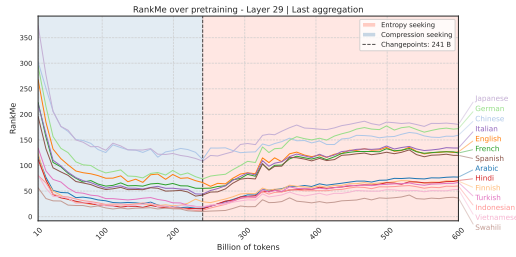


Figure 1: Last-layer RankMe evolution of FuxiTranyu.

4.2 Layer-Depth Geometry

Figure 2 overlays each language’s mean RankMe profile with depth phases identified jointly from all four spectral metrics (Appendix B.4): FuxiTranyu traverses four phases and Apertus three, mid-depth *compression* (layers 15–19) being the sole structural difference between the models. Cross-language spread grows with depth in both models (H2) but follows each model’s phase structure. The hierarchy is stable: per-checkpoint profiles closely track each language’s mean profile (H1), and the ranking is fixed within $\sim 7\text{--}9\%$ of training (Appendix B.1). It is also largely architecture-driven, grouping languages by script family and morphological complexity with >0.85 per-layer rank agreement between the two models despite a $25\times$ difference in training tokens (H3, Appendix B.3). The remaining metrics corroborate this picture and expose signatures invisible to RankMe, e.g. a final-layer rank bottleneck in Apertus (Appendix B.5). Because the hierarchy is fixed so early, first-checkpoint RankMe already predicts downstream performance (§4.4).

4.3 Data Metrics as Geometry Predictors

For the results of this section, all analyses are conducted using all selected languages, as well as excluding Chinese and Japanese, whose fertility is computed on a different unit basis and may otherwise dominate the signal.

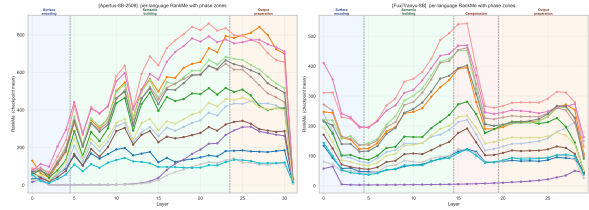


Figure 2: Per-language RankMe by layer (checkpoint mean) with geometric phase zones overlaid (colors as in Figure 1). Left: Apertus (3 phases). Right: FuxiTranyu (4 phases, incl. compression at layers 14–19).

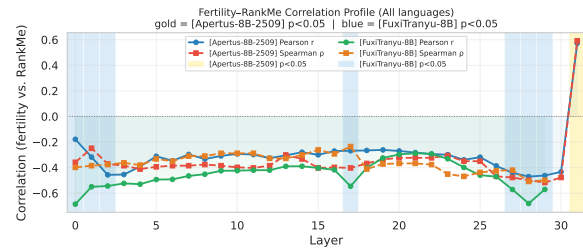


Figure 3: Fertility–RankMe correlations at last checkpoint. Gold and blue bands indicate statistical significance for Apertus and FuxiTranyu respectively.

Training data distribution (FuxiTranyu). Figure 23 (appendix) shows the layer-by-layer correlation between training data share and RankMe. The signal follows a depth-dependent pattern, positive near the embedding and output layers, and negative in the middle. Yet, statistical significance is never reached. The result shows how in the middle layers, whose representations are far from the tokenizer space, more represented languages in the training set have more compressed representations.

Fertility– and coverage–RankMe correlation.

Figure 3 shows layer-by-layer Pearson r and Spearman ρ between fertility and RankMe at the final checkpoint, with weak mostly non-significant correlations in both models. A consistently negative correlation is present in both models throughout the whole training (see Figure 24). These results suggest that languages with lower fertility require more dimensions per token, perhaps as each token spans a larger lexical unit and carries more semantic content. Vocabulary coverage mirrors these findings with inverted sign, indicating that languages that use a larger portion of the input vocabulary are less compressed in the embedding space. Overall, we find that tokenizer fertility and vocabulary coverage can influence the RankMe evolution, although statistical significance is rarely obtained. Refer to appendix C for the complete results.

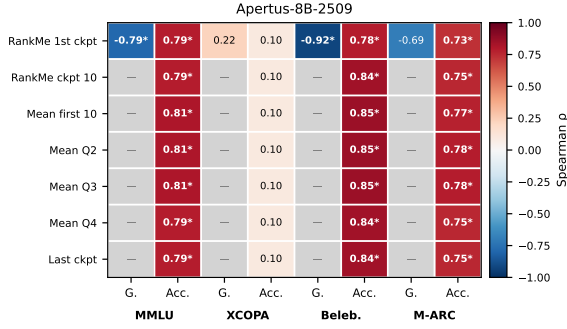


Figure 4: Spearman ρ between RankMe predictors (rows) and downstream outcomes (columns) for Apertus. Two columns per task: G. = grokking onset (billions of tokens); Acc. = peak accuracy. Beleb. = Belebele. Q2–Q4: mean RankMe over the 25–50%, 50–75%, and 75–100% of training tokens. Bold *: $p < 0.05$.

4.4 RankMe and Monolingual Performance

Figure 4 summarises Spearman ρ for all predictor–outcome pairs for Apertus; the corresponding FuxiTranyu heatmap is Figure 38 in the appendix. For **Apertus**, RankMe strongly predicts both grokking onset and peak accuracy on m-MMLU, Belebele, and M-ARC: all 7 predictors are significant for peak accuracy (Sp $\rho = +0.73$ – 0.85 , $p < 0.025$; higher RankMe $\Rightarrow$ higher accuracy), and first-checkpoint RankMe negatively predicts grokking onset (higher RankMe $\Rightarrow$ earlier grokking). XCOPA is the exception: outcomes cluster tightly (0.70–0.78), leaving no cross-language variance to detect. For **FuxiTranyu**, m-MMLU and Belebele accuracy stays near random chance throughout training, producing no detectable signal. The sole cross-model replication is **M-ARC**: RankMe at the first checkpoint predicts peak accuracy in both models (FuxiTranyu: Pearson $r = +0.82$, $p = 0.007$; Apertus: Sp $\rho = +0.73$, $p = 0.025$). Per-language plots for all benchmarks and models are in the appendix (Figures 39–46).

Figure 47 (Appendix E.3) shows the per-checkpoint Spearman correlation between the per-language RankMe ranking and accuracy ranking for both models. On m-MMLU, Belebele, and M-ARC, the RankMe language ranking tracks the accuracy ranking throughout most of training on Apertus. XCOPA shows no significant checkpoints. A complementary AlphaReQ analysis confirms these findings with flipped signs, consistent with its inverse relationship to RankMe (Garrido et al., 2023); see Figures 48 and 49 in the appendix.

4.5 RankMe and Cross-Lingual Transfer

Figure 5 shows the Spearman r between geometry predictors at checkpoint C and transfer score $\tau(s, t)$ at checkpoint $C+T$, pooling all language pairs. FuxiTranyu shows no meaningful correlation due to low variance across language pairs (see D.2.2). For Apertus, the RankMe difference between source and target (src–tgt) achieves a strong negative correlation (up to $|r| = 0.80$ at individual checkpoints, see D.2.1), persisting up to $T = 30$ checkpoints ahead ($\sim 10T$ tokens). It outperforms individual RankMe values (src and tgt) and the absolute difference ($|\text{src} - \text{tgt}|$), showing that transfer depends on the *direction* of the effective-dimensionality gap. In particular, transfer is more effective from low- to high-RankMe languages than the reverse. Together, the results suggest that RankMe can serve as an early predictor of downstream cross-lingual transfer. We verify these findings across different settings in D.2.3.

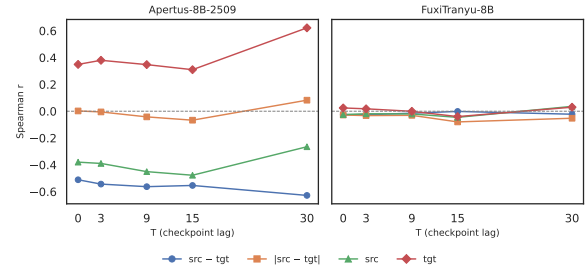


Figure 5: Spearman r between geometry predictors and transfer score when predicting T checkpoints ahead. Filled markers indicate significance ($p < 0.05$).

5 Ethical considerations

This work aims to reduce language disparities in multilingual LLMs by diagnosing representation imbalances early in training. However, our analysis covers only 14 languages, skewing toward high-resource ones, so speakers of unrepresented languages may not benefit equally.

6 Conclusion

We showed that spectral geometry evolves uniformly across languages and across models with very different training scales. Moreover, we found that it predicts both monolingual and cross-lingual performance, making it an early, interpretable signal of multilingual capabilities. However, our findings are preliminary: we only used two models and 14 languages, and most of our performance results apply to Apertus only.

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A Geometric Evolution of Languages

A.1 Languages

We compute RankMe for 14 languages from diverse families and with varying levels of representation in the models’ training data. Table 1 lists the languages, families, and branches.

Language	Family	Branch
English	Indo-European	Germanic
Chinese	Sino-Tibetan	Sinitic
German	Indo-European	Germanic
French	Indo-European	Romance
Spanish	Indo-European	Romance
Italian	Indo-European	Romance
Arabic	Afro-Asiatic	Semitic
Vietnamese	Austroasiatic	Mon-Khmer
Japanese	Japonic	Japanese
Turkish	Turkic	Oghuz
Finnish	Uralic	Finnic
Hindi	Indo-European	Indo-Iranian
Indonesian	Austronesian	Malayo-Polynesian
Swahili	Niger-Congo	Benue-Congo
Total	9 families	11 branches

Table 1: Language families and branches.

A.2 Last-Layer RankMe Evolution

Figure 6 shows the last-layer RankMe trajectory for all 14 languages across the full pretraining Apertus respectively. The higher training token count (15T vs. 600B for FuxiTranyu) allows analysis of geometric co-evolution over longer training regimes. Similarly to FuxiTranyu, we find that all languages mostly display the same evolution. However, in this model, we also observe a third phase characterized by slight compression at the end of pretraining.

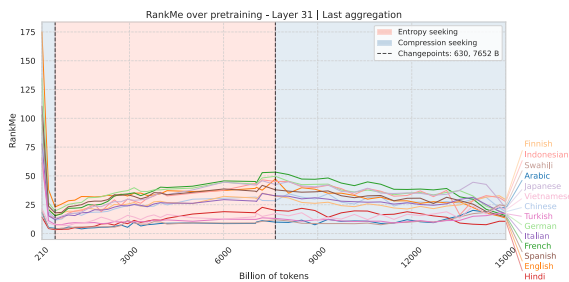


Figure 6: Last-layer RankMe evolution of Apertus.

A.3 Laws of Geometric Evolution

In this section, we characterize the geometric evolution of the 14 languages in our two models by fitting shared parametric laws to their RankMe training curves. We focus on RankMe at the last layer.

We first analyze per-segment normalized curves to show that languages share the same shape regardless of magnitude, then fit piecewise models that assume a common functional form with only a few per-language parameters (α, β). This yields a compact, interpretable description of how languages differ geometrically during training.

A.3.1 FuxiTranyu

We begin by normalizing curves by applying min-max scaling independently within each phase (per-segment normalization), mapping each language’s curve to $[0, 1]$ per phase to remove magnitude differences while preserving shape.

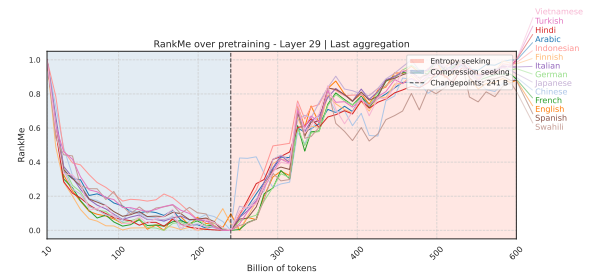


Figure 7: Normalized last-layer RankMe evolution of FuxiTranyu. Min-max scaling values per phase shows that languages exhibit the same geometric phases.

The per-segment normalized RankMe curves of all 14 languages exhibit the same two-phase shape: an initial power-law decay followed by an exponential recovery, differing across languages only in overall magnitude and offset (Figure 7). This motivates a piecewise model of the form

$$\hat{R}(t) = \begin{cases} \alpha + A t^{-\gamma} & t \leq t_1, \\ \alpha + A P_1 + C \left(1 - e^{-(t-t_1)/\lambda}\right) & t > t_1, \end{cases} \quad (1)$$

where $P_1 = t_1^{-\gamma}$ ensures continuity at the change-point t_1 . Here α is the asymptotic floor of the compression phase, A controls the per-language amplitude of the power-law decay, and C controls the per-language amplitude of the exponential recovery; the decay exponent γ and the recovery timescale λ are shared across all languages. Fitting this model to FuxiTranyu-8B yields $R^2 \geq 0.90$ for all 14 languages (minimum: Swahili at 0.91), confirming that γ and λ are universal while only α , A , and C vary by language (see Figure 9a).

A.3.2 Apertus

We start again by applying per-segment normalization of the curves (Figure 8). We find that all

languages share the same first two phases: an initial power-law decay, followed by an exponential recovery toward a plateau. In the third phase, however, the languages diverge: four languages (Arabic, Hindi, Swahili, Vietnamese) exhibit a final RankMe expansion, while the remaining ten compress. We therefore model each language with two mutually exclusive phase-3 branches, defined through the unified basis function

$$b(t, \lambda) = \left(e^{(t-t_2)/\lambda} - 1 \right) \mathbf{1}[t > t_2], \quad (2)$$

which saturates in $(-1, 0]$ for $\lambda < 0$ (decay) and grows unboundedly for $\lambda > 0$ (growth). Setting $P_1 = t_1^{-\gamma}$, $e_2(t) = 1 - e^{-(t-t_1)/\lambda_2}$, and $Q_2 = e_2(t_2)$, the full model is

$$\hat{R}(t) = \begin{cases} \alpha + A t^{-\gamma} & t \leq t_1, \\ \alpha + A P_1 + C_2 e_2(t) & t_1 < t \leq t_2, \\ \alpha + A P_1 + C_2 Q_2 + C_{3,d} b(t, -\lambda_{3,d}) & t > t_2, \text{ decay}, \\ \alpha + A P_1 + C_2 Q_2 + C_{3,g} b(t, +\lambda_{3,g}) & t > t_2, \text{ growth}, \end{cases} \quad (3)$$

where continuity at t_1 and t_2 is enforced by P_1 and Q_2 . Per-language parameters are α (asymptotic floor of the compression phase), A (phase-1 decay amplitude), C_2 (phase-2 recovery amplitude), and either $C_{3,d} \geq 0$ or $C_{3,g} \geq 0$ (phase-3 amplitude, exactly one active per language). The shared parameters are the power-law exponent γ , the phase-2 recovery timescale λ_2 , and the separate phase-3 timescales $\lambda_{3,d}$ and $\lambda_{3,g}$ for the two branches. This model achieves $R^2 \geq 0.86$ for all languages except Vietnamese ($R^2 = 0.68$), with most languages above 0.90 (see Figure 9b).

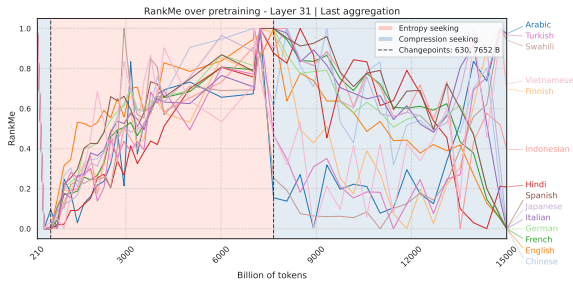
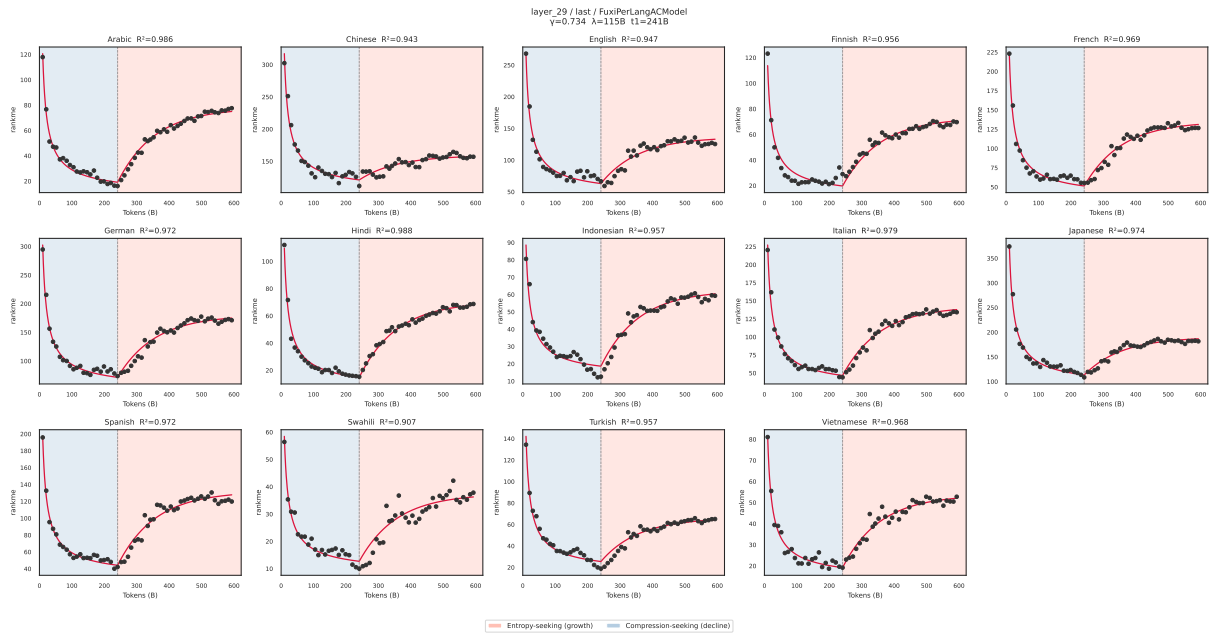
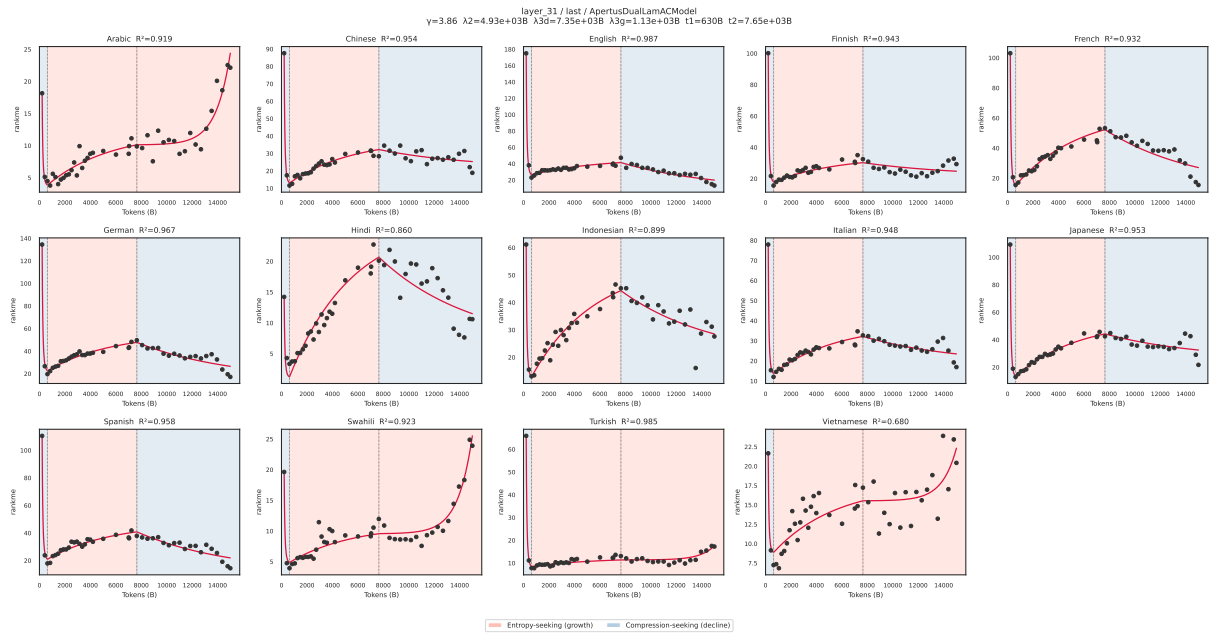


Figure 8: Normalized last-layer RankMe evolution of Apertus. Min-max scaling values per phase shows that languages exhibit the same geometric phases.



(a) RankMe trajectories for FuxiTranyu fitted with a two-phase piecewise model. Each panel shows one language; dots are measured RankMe scores at training checkpoints, the crimson curve is the fitted model, and the dashed line marks the changepoint. Red and blue shading indicate entropy-seeking (rising) and compression-seeking (falling) phases, respectively.



(b) RankMe trajectories for Apertus fitted with a three-phase piecewise model. Each panel shows one language; dots are measured RankMe scores at training checkpoints, the crimson curve is the fitted model, and dashed lines mark the two changepoints. Red and blue shading indicate entropy-seeking (rising) and compression-seeking (falling) phases, respectively.

Figure 9: RankMe trajectories for FuxiTranyu (top) and Apertus (bottom) fitted with piecewise models.

B Layer-Depth Geometry: Detailed Results

This section expands Section 4.2. Appendix B.1 quantifies profile stability and convergence (H1); Appendix B.2 covers cross-language stratification by depth (H2); Appendix B.3 examines the origin of the language hierarchy (H3); Appendix B.4 describes how the depth phases of Figure 2 are identified; Appendices B.5 and B.6 extend the analysis to the three remaining spectral metrics and study how the four metrics relate to one another.

B.1 Profile Stability and Convergence (H1)

Stability. For each language and checkpoint we measure the cosine similarity between its per-layer RankMe profile and that language’s checkpoint-mean profile. Figure 10 shows per-checkpoint profiles (faint lines) closely tracking the checkpoint-mean profile (bold) throughout training: mean similarity is 0.99 for Apertus and 0.91 for FuxiTranyu (excluding Hindi, which is degenerate in FuxiTranyu; see Appendix B.3). Training therefore rescales RankMe magnitude per layer without reshaping the depth profile. The FuxiTranyu minimum (0.59) falls in the earliest checkpoints of languages that later develop the mid-layer compression collapse: the profile shape is not fully locked at the start, but stabilizes within roughly the first quarter of training.

The ranking is fixed early. Figure 11 compares the mean layer profile over the first 5 checkpoints against the last 5. The two sets of profiles are nearly indistinguishable in shape, and the cross-language ranking at the representative layer already agrees at Spearman $\rho > 0.90$ ($p < 0.001$) – even though the first 5 checkpoints cover only $\sim 9\%$ of FuxiTranyu’s training (10–52B tokens) and $\sim 7\%$ of Apertus’s (210B–1.05T of 15T). The dominant change over training is a near-uniform upward scaling of RankMe; the rank *order* of languages is set almost from the first gradient steps, suggesting it reflects architectural inductive biases (tokenizer coverage, embedding geometry) more than the progressive accumulation of training data.

Convergence. Figure 12 tracks, per checkpoint, the cosine similarity between each language’s layer profile and its final-checkpoint profile. FuxiTranyu profiles converge above 0.99 within $\sim 20\%$ of training; Apertus profiles start at or above ~ 0.85 already at the first public checkpoint (210B tokens,

i.e. 1.4% of its 15T budget), indicating that the rapid shape-formation regime ends very early. One apparent exception is instructive: Hindi in FuxiTranyu looks maximally stable only because its profile is near zero at every layer and checkpoint – the model never develops meaningful Hindi representations, so the shape is trivially constant. This early convergence is what makes first-checkpoint RankMe a usable predictor of downstream performance (§4.4).

B.2 Cross-Language Stratification (H2)

Figure 13 plots the across-language standard deviation of RankMe at each layer (checkpoint mean). Spread grows with depth in both models – early layers are nearly language-agnostic and the strongest language-specific geometry sits in the middle-to-late layers – but the two architectures diverge inside this trend, and the divergence follows their phase structure (Appendix B.4). FuxiTranyu’s spread peaks at layer 15, exactly at the boundary between semantic building and compression, then collapses as every language is squeezed through the compression zone and only partially re-stratifies afterwards. Apertus’s spread instead grows continuously and peaks at layer 21, near the end of its long semantic-building phase, before a modest decline during output preparation. Cross-language stratification is therefore not an incidental property of depth: it is built up during semantic building and is then either destroyed (FuxiTranyu) or preserved (Apertus) by what follows.

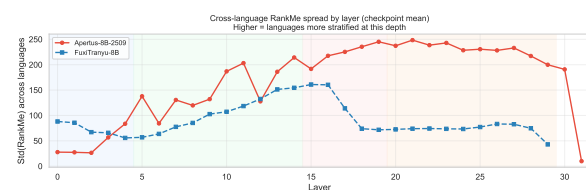


Figure 13: Cross-language standard deviation of RankMe at each layer (H2), both models. FuxiTranyu peaks at layer 15 (the semantic-building/compression boundary); Apertus at layer 21.

B.3 Origin of the Language Hierarchy (H3)

Clustering. Figure 14 shows Ward-linkage agglomerative clustering of languages on their checkpoint-mean layer-profile vectors (Euclidean distance on raw, un-normalized profiles, so both magnitude and shape contribute). In FuxiTranyu the clusters track training-data share and script more than strict linguistic family: {German,

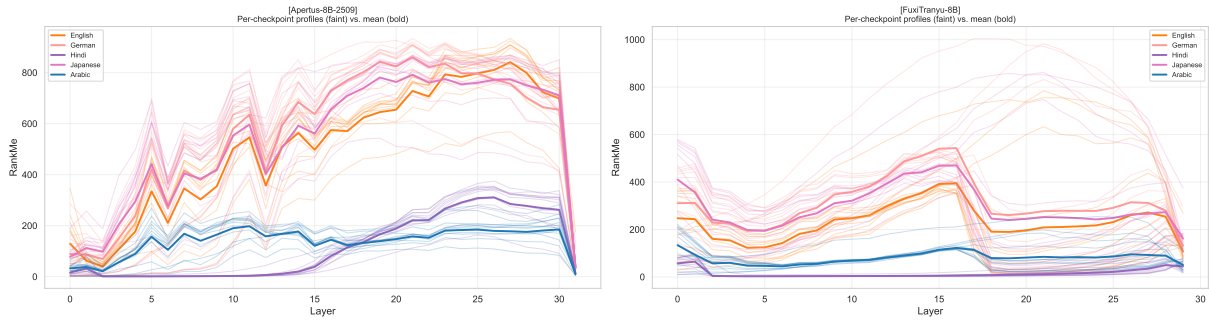


Figure 10: Per-checkpoint layer profiles (faint lines) vs. checkpoint-mean profile (bold), both models. Profiles closely track the mean throughout training (H1); training shifts magnitude, not shape.

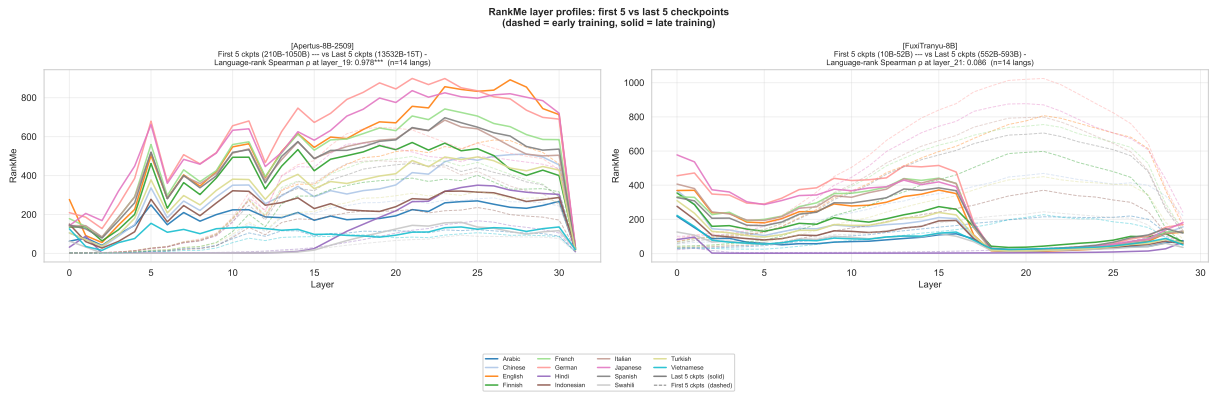


Figure 11: Mean layer profile averaged over the first 5 vs. last 5 training checkpoints, both models. The language ranking from the two windows already agrees at $\rho > 0.90$, showing that the cross-language hierarchy is fixed early in training.

Japanese} (high early-layer rank, strong deep-layer collapse), {English, French, Italian, Spanish} (moderate, steadily rising profiles), {Arabic, Indonesian, Swahili, Vietnamese} (flatter, low-resource-like profiles), and Hindi as a singleton. Hindi’s RankMe averages 12 at layer 21 against English’s 209 – a 17-fold deficit that is uniform across every layer and checkpoint – indicating that FuxiTranyu never builds usable Hindi representations, most plausibly due to a minimal Hindi share in its corpus. In Apertus no language is isolated: Hindi remains the lowest (166 at layer 19, vs. 645 for English) but joins a low-resource cluster with Swahili and Vietnamese, and German and Japanese fold into the European group. As training data grows larger and more balanced ($25\times$ more tokens), the groupings become more typologically coherent.

Cross-model consistency. Figure 15 shows the per-layer Spearman ρ between the Apertus and FuxiTranyu language rankings: $\rho > 0.85$ at every one of the 30 comparable layers (all $p < 0.05$), despite the $25\times$ difference in training tokens. Agreement is uniformly high in every depth phase (mean

$\rho = 0.92/0.95/0.95/0.90$ in surface encoding / semantic building / compression / output preparation): even FuxiTranyu’s compression zone, which dramatically shrinks absolute RankMe, does not disturb the relative ordering of languages. The hierarchy is structural, not an artifact of scale or of either training run.

Which languages hold their position? We also correlate, per language, its rank trajectory across the 30 shared layers in one model with the same trajectory in the other. German ($\rho = 0.57$, $p < 0.001$) and Japanese ($\rho = 0.45$, $p < 0.05$) are the most universally positioned languages. Vietnamese is the lone inversion ($\rho = -0.43$, $p < 0.05$): its absolute RankMe is almost identical in the two models (93 vs. 92 at the representative layers), but European languages compress far more strongly than Vietnamese inside FuxiTranyu’s mid-layer collapse, which pulls Vietnamese’s *relative* rank up precisely at the depths where it is low in Apertus. The inversion is thus an artifact of the compression asymmetry rather than a genuine representational difference.

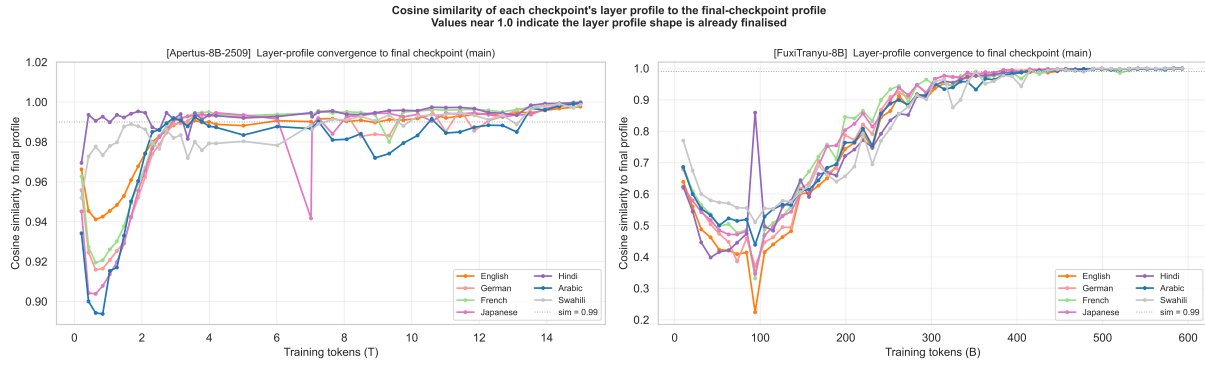


Figure 12: Cosine similarity of each checkpoint’s layer profile to the final-checkpoint profile, for 7 representative languages, both models. FuxiTranyu converges within $\sim 20\%$ of training; Apertus is essentially converged at its first public checkpoint.

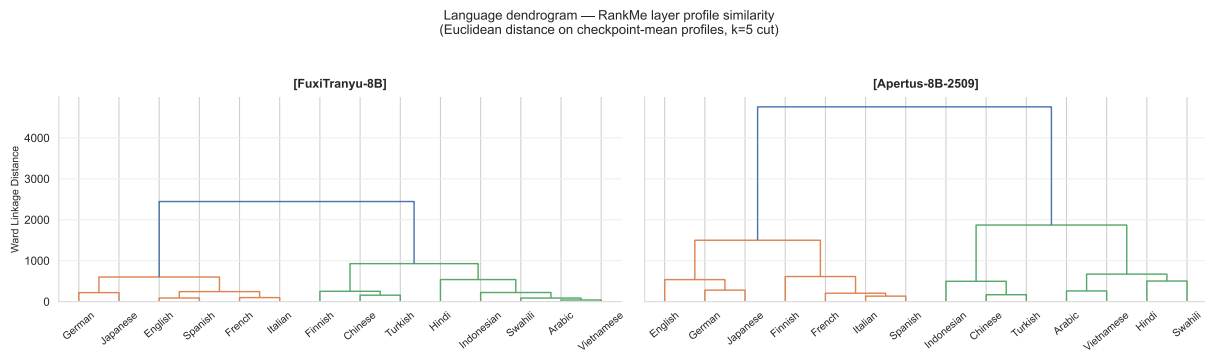


Figure 14: Ward-linkage clustering of languages on mean layer-profile vectors (H3), both models. FuxiTranyu groups languages by training-data share and script, with Hindi as a degenerate singleton; Apertus clusters are more typologically coherent, with no isolated outlier.

B.4 Identifying Depth Phases

The phase zones of Figure 2 are derived from all four spectral metrics jointly. For each model we min–max normalize each metric’s checkpoint-mean layer profile to $[0, 1]$ (inverting AlphaReQ so that higher always means richer geometry), average the four normalized profiles into a combined signal, smooth it with a Gaussian of $\sigma = 1$ layer, and place phase boundaries at its inflection points (Figure 16). This yields boundaries at layers 5 and 24 for Apertus and at layers 5, 15, and 20 for FuxiTranyu. No formal change-point detection is applied, so the boundaries should be read as indicative rather than precise.

The result is three phases for Apertus – surface encoding (layers 0–4), semantic building (5–23), output preparation (24–31) – and four for FuxiTranyu – surface encoding (0–4), semantic building (5–14), *compression* (15–19), output preparation (20–29). Compression is the sole structural addition, and it reshapes everything downstream: Apertus spends 19 layers (60% of its depth) in

semantic building, FuxiTranyu only 10, and FuxiTranyu’s combined signal never recovers its pre-compression level afterwards (0.66 vs. 0.78). What looks like a recovery after layer 19 is better read as a degraded descent toward the LM head – functionally the same output-preparation role as Apertus layers 24–31, but entered from a compressed state. Figure 2 confirms that this phase structure is not an averaging artifact: every individual language rises through semantic building, collapses inside FuxiTranyu’s compression zone, and declines during output preparation, all while maintaining its rank position.

B.5 Beyond RankMe: PR, AlphaReQ, and Top-K Variance

We repeat the layer-profile analysis for the remaining spectral metrics. They corroborate the RankMe picture where the metrics overlap, and expose three signatures that RankMe cannot see.

Participation Ratio. In Apertus (Figure 17, left), PR grows continuously to a peak near layer 21

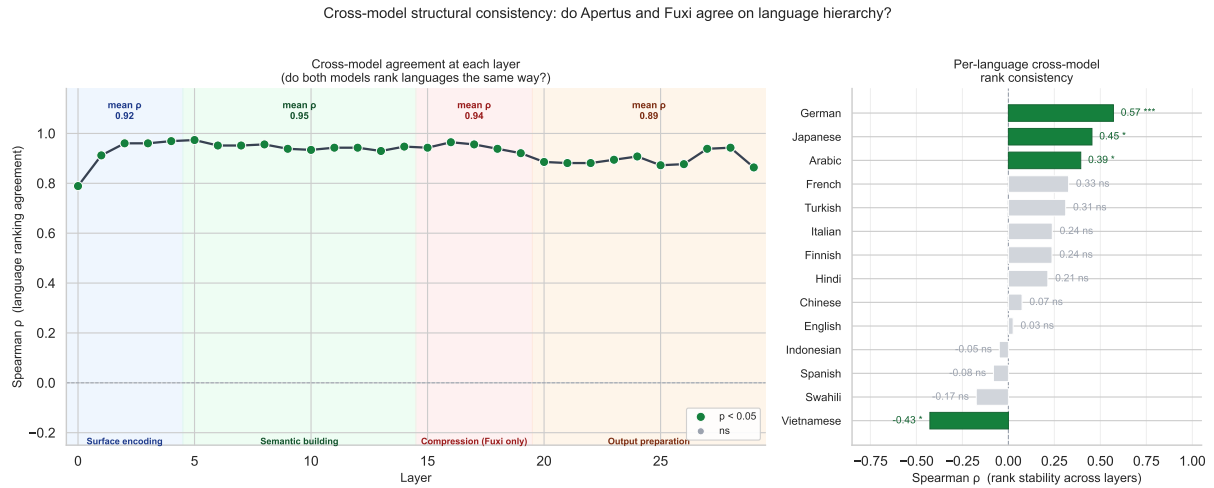


Figure 15: Per-layer Spearman ρ between Apertus and FuxiTranyu language rankings (left) and per-language rank consistency across depth (right). The cross-language hierarchy agrees at $\rho > 0.85$ at every layer; Vietnamese is the only language whose relative position inverts between models.

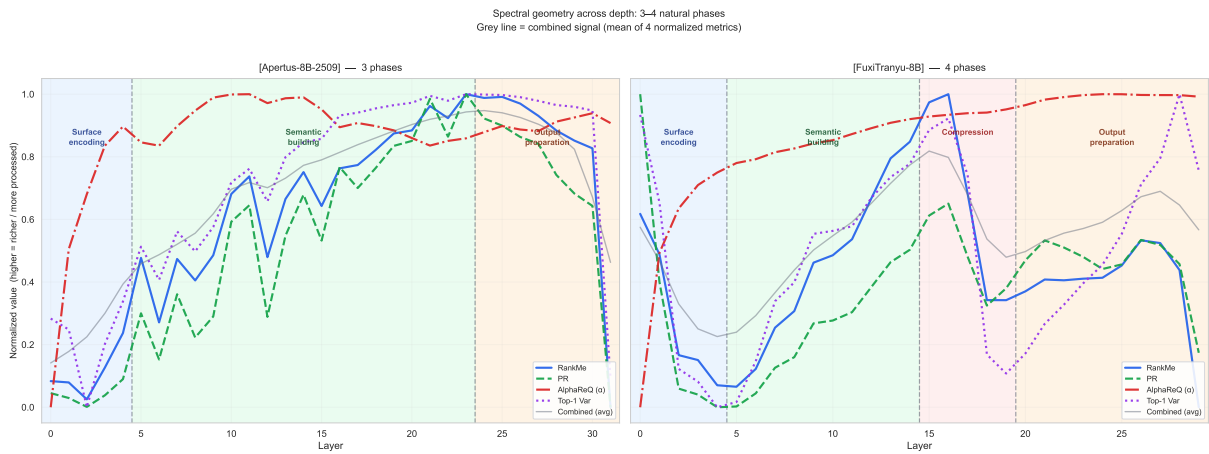


Figure 16: Depth-phase identification. Per-metric normalized layer profiles (colored), their combined signal (grey), and the resulting phase zones for Apertus (left, 3 phases) and FuxiTranyu (right, 4 phases). Boundaries are placed at inflection points of the smoothed combined signal.

(Japanese reaches ~ 120) and its cross-language spread peaks at the same depth – mirroring RankMe. FuxiTranyu (right) instead shows an embedding-layer spike that RankMe misses entirely: Japanese enters at PR ≈ 70 at layer 0 and collapses to near baseline by layer 3, and cross-language PR spread consequently peaks at layer 0. The embedding assigns many participating directions to morphologically complex languages, but the first transformer blocks immediately suppress them: the two models disagree about *which depth* carries language-specific information.

AlphaReQ. α decays monotonically with depth in both models (FuxiTranyu $\sim 2.2 \rightarrow 0.65$; Apertus $\sim 1.2 \rightarrow 0.65$; Figure 18): earlier layers have steeper, more concentrated singular-value spectra.

The notable asymmetry is in stratification: in FuxiTranyu, α spread across languages peaks at layer 0 (std ≈ 0.27) and decays to near zero with depth – the *opposite* of RankMe – meaning its transformer blocks erase language-specific spectral structure rather than building on it. Apertus keeps α spread roughly flat (≈ 0.10) at all depths.

Top- K Variance. The Top- K family (Figure 19) exposes an Apertus-specific final-layer bottleneck: at layer 31 the top singular component captures nearly all activation variance (top-1 fraction $\rightarrow 1.0$ for English). The bottleneck is present from the earliest checkpoints, marking it as an architectural property – plausibly the LM head forcing a low-dimensional output projection – yet it is invisible to both RankMe and PR. In FuxiTranyu, the small- K

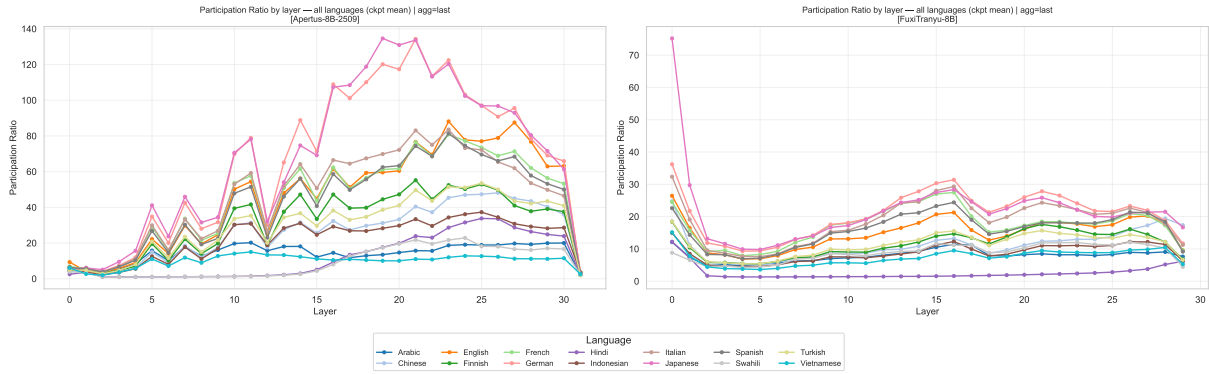


Figure 17: Participation Ratio by layer (checkpoint mean), per language. Apertus (left) grows continuously to a peak near layer 21; FuxiTranyu (right) shows an embedding-layer spike for morphologically complex languages (Japanese PR ≈ 70 at layer 0) that the first transformer blocks suppress.

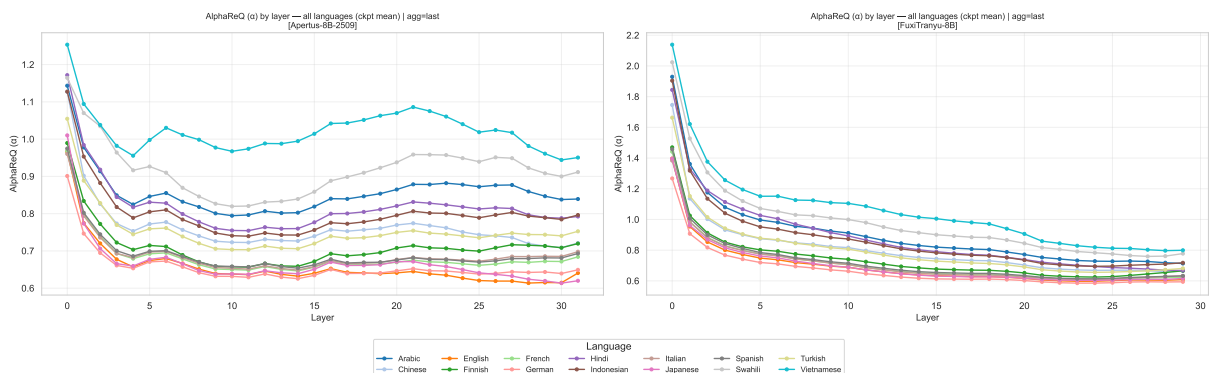


Figure 18: AlphaReQ (α) by layer (checkpoint mean), per language. α decays monotonically with depth in both models; cross-language α spread peaks at the embedding in FuxiTranyu and decays with depth, inverting the RankMe stratification pattern.

curves bump upward at layers 14–20, the spectral counterpart of the compression collapse: when effective rank drops, variance redistributes into fewer dominant components. Per-language extremes line up with the rest of the analysis: Hindi in FuxiTranyu has mean top-1 variance 0.72 (vs. ~ 0.30 for most languages), i.e. its activations collapse onto a single direction, while Japanese in Apertus sits at the opposite extreme (0.19), consistent with its high PR and RankMe. Finally, the cross-language spread of top-1 variance reveals a transient, Apertus-only differentiation window: languages are spectrally indistinguishable at layer 0, briefly diverge at layers 5–10 (spread ≈ 0.25), and re-converge by layer 12 – an early-depth phase with no counterpart in RankMe, PR, or α .

B.6 Cross-Metric Correlation Structure

Are the four metrics redundant? Figure 20 correlates the four checkpoint-mean layer profiles pairwise. In Apertus, RankMe, PR, and Top-1 Variance form a near-perfectly coupled triplet ($|\rho| \geq 0.96$,

all $p < 0.001$): they measure eigenvalue spread from three angles and agree almost exactly. AlphaReQ is the outlier, only moderately correlated with RankMe ($\rho = -0.45$) and PR ($\rho = -0.38$), confirming it captures a distinct spectral property. In FuxiTranyu this coupling breaks down: RankMe–PR drops to $\rho = 0.74$, and AlphaReQ decouples from every other metric ($|\rho| \leq 0.11$, all n.s.). The decoupling quantifies the compression effect in a single number – α continues its monotone decay while RankMe collapses and recovers, destroying the correlation structure that holds in the data-richer model.

Which metric drives each phase? Figure 21 correlates each metric with layer index *within* each phase. During surface encoding, AlphaReQ is the only significant metric in both models ($\rho \approx -0.93$ to -1.00): it moves before RankMe and PR begin to react, making it a leading indicator of early representational structure. During semantic building the roles flip: RankMe, PR, and Top-1 Variance

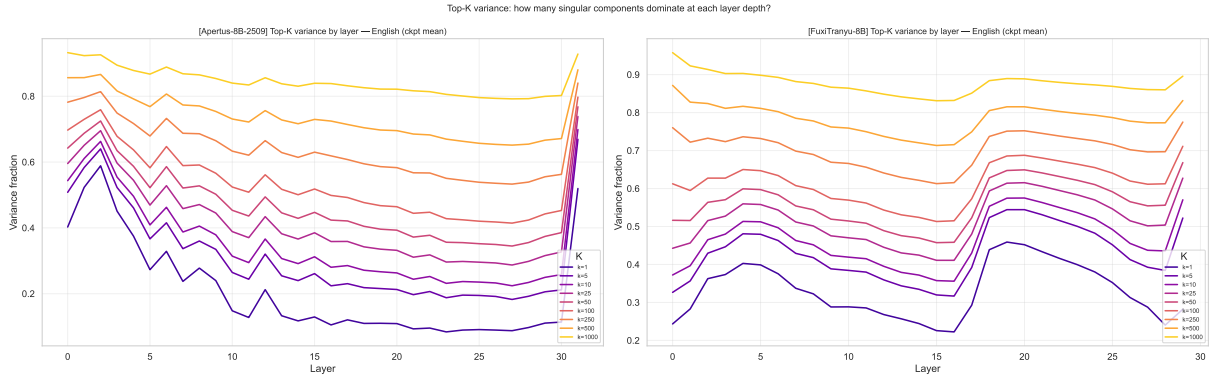


Figure 19: Top- K variance fraction by layer for English, $K \in \{1, \dots, 1000\}$, both models. Apertus (left) shows the final-layer bottleneck at layer 31 (top-1 $\rightarrow 1.0$); FuxiTranyu (right) shows the small- K concentration bump at layers 14–20 that mirrors its compression phase.

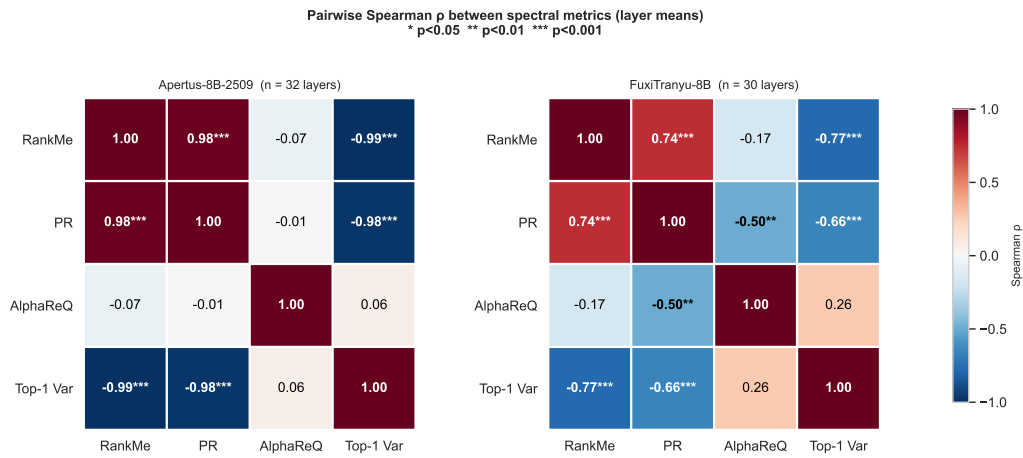


Figure 20: Pairwise Spearman ρ between the four spectral metrics' layer profiles. RankMe, PR, and Top-1 Variance form a tightly coupled triplet in Apertus (left) but decouple in FuxiTranyu (right), where AlphaReQ becomes independent of every other metric ($|\rho| \leq 0.11$, n.s.).

are all strongly significant ($|\rho| \geq 0.88$, $p < 0.001$) while AlphaReQ, already flattened, drops to non-significance. In FuxiTranyu's compression phase all four metrics are simultaneously significant, reflecting the sharp, coordinated nature of the transition. The phase decomposition of Appendix B.4 therefore does not merely segment a single curve: each phase has a distinct spectral driver.

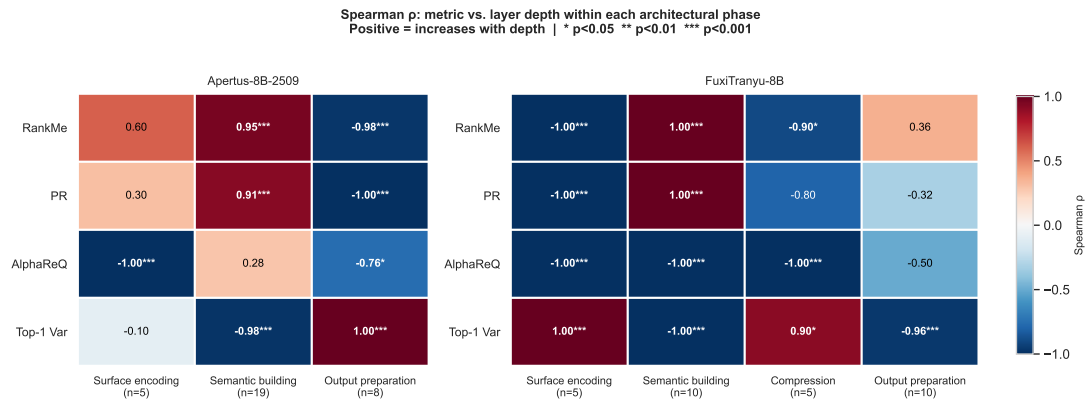


Figure 21: Spearman ρ between each spectral metric and layer index, computed within each depth phase (Apertus left, FuxiTranyu right). AlphaReQ is the only significant metric in surface encoding; RankMe, PR, and Top-1 Variance dominate semantic building; all four co-move in FuxiTranyu’s compression phase. *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.

C Tokenizer metrics as predictors of Representation Geometry

C.1 Training distribution vs. RankMe correlation

Figure 23 shows the correlation between the training distribution and the RankMe scores per-language, for FuxiTranyu. Apertus’ full training distribution per-language is not available, so the model is excluded from this study. The results show that the correlation is positive at the extremes of the model, where representations are more strictly linked to surface-level language features, and becomes negative in the middle layers, where representations are more abstract. This suggests that the most frequently seen languages during training are more efficiently compressed in the middle layers of the model, while they use more embedding dimensions in the final layers, better preparing for the final projection onto the vocabulary space.

C.2 Fertility–RankMe and Coverage–RankMe correlation

Figure 25 shows the correlation between fertility and RankMe ((a) and (b)), as well as between coverage and RankMe ((c) and (d)), calculated for the last layer of the last checkpoint of each model. The results show a statistical significance for the Fertility–RankMe correlation in the first and last three layers of Fuxi. The correlation is negative across all layers, indicating that more fertile languages (worse tokenization efficiency) have lower RankMe scores. The only exception occurs in the last layer of Apertus, where the correlation suddenly becomes positive and statistically significant. This observation is hard to justify: it could happen because the last layer has to produce the final embedding used for the vocabulary projection, and the fact that we do not observe the same behavior in FuxiTranyu is probably due to architectural choices or to the training compute gap between the two models.

C.3 Correlation heatmap for all checkpoints, all layers

Figure 24 shows Spearman ρ between tokenization metrics and RankMe at every (checkpoint, layer) pair, across the full training trajectory of both models.

Fertility–RankMe. The heatmap is predominantly blue for both models, indicating a consis-

tently negative correlation: languages with higher fertility tend to exhibit lower RankMe throughout training. The effect is not uniform in depth: it is strongest near the input and output layers, where representations are most directly tied to the token surface, and weakens in the middle of the network. Despite the consistent direction, most individual cells do not reach statistical significance ($p < 0.05$), so the pattern should be read as a directional tendency rather than a robust quantitative law.

Coverage–RankMe. The coverage heatmap mirrors the fertility result with inverted sign: higher vocabulary coverage is associated with higher RankMe. This is expected, since coverage and fertility are inversely related, meaning that a language whose text is encoded into fewer, more familiar tokens will activate a broader share of the vocabulary and achieve higher representational diversity. The symmetry between the two heatmaps thus reflects the same underlying phenomenon measured from complementary angles.

Subcaptions identify what’s shown; the main caption owns the color-coding interpretation. The after (a) and (b) prevents a line break between the label and the text.

C.4 Fertility and RankMe evolution

Figure 26a shows the per-language RankMe evolution during training for both models, with lines colored by fertility (bright = low, dark = high). In neither model does the fertility ordering of languages map onto a consistent RankMe ordering: brighter and darker lines interleave throughout training, with no persistent separation between tiers. This confirms visually that tokenizer efficiency does not impose a reliable structure on the evolution of representational geometry.

Figure 26b repeats the same analysis with lines colored by vocabulary coverage (dark = low, bright = high). The picture is identical in kind: languages with similar coverage values do not cluster together in RankMe space, and no stable ordering emerges across training. This double null result suggests that neither dimension of tokenization quality (fragmentation density nor vocabulary breadth) is a strong predictor of representation geometry at the macro level.

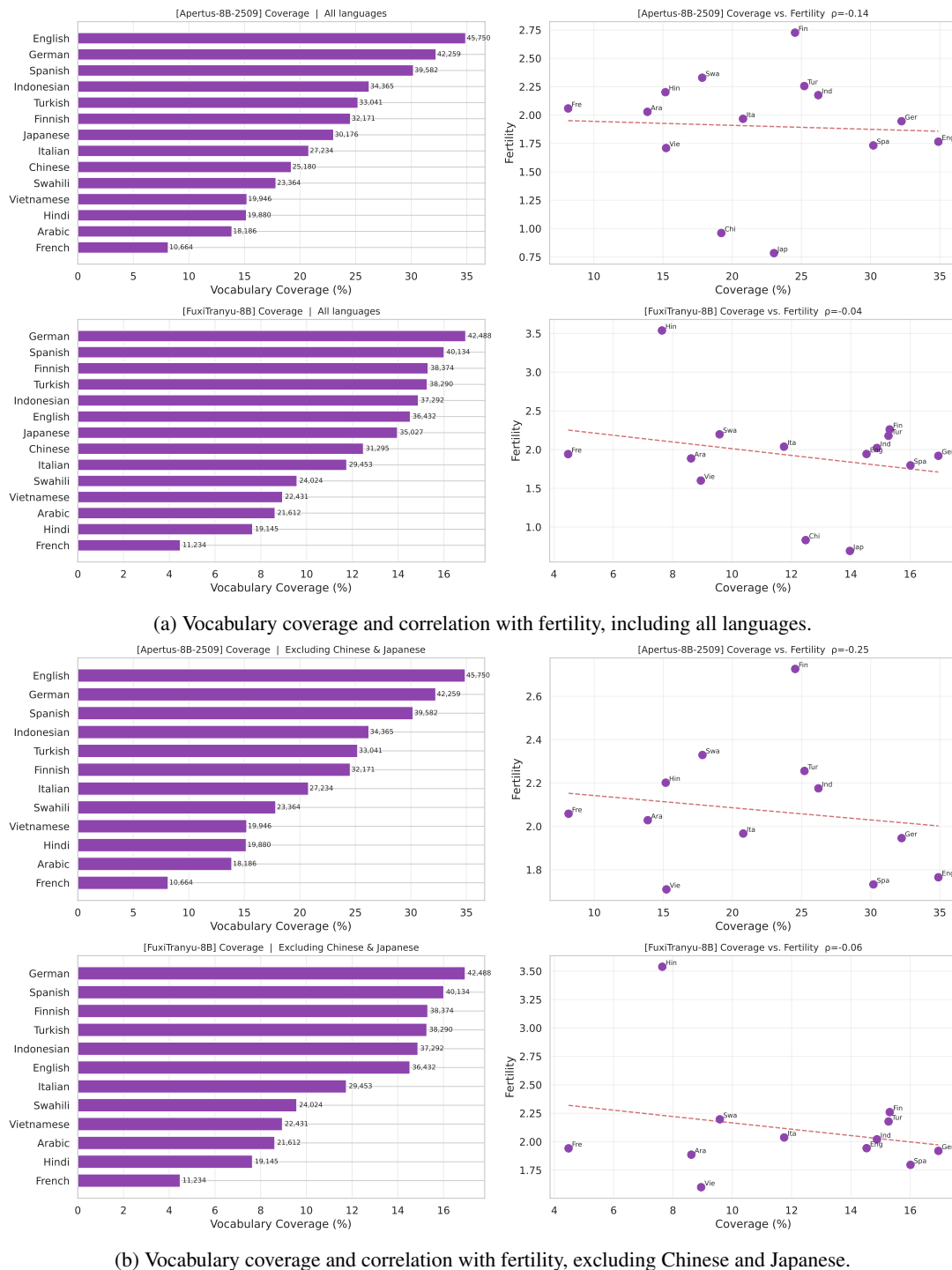
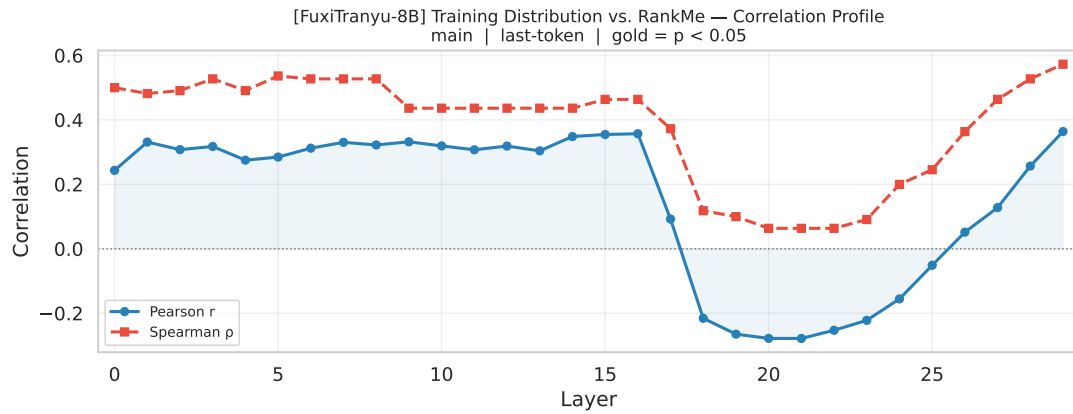
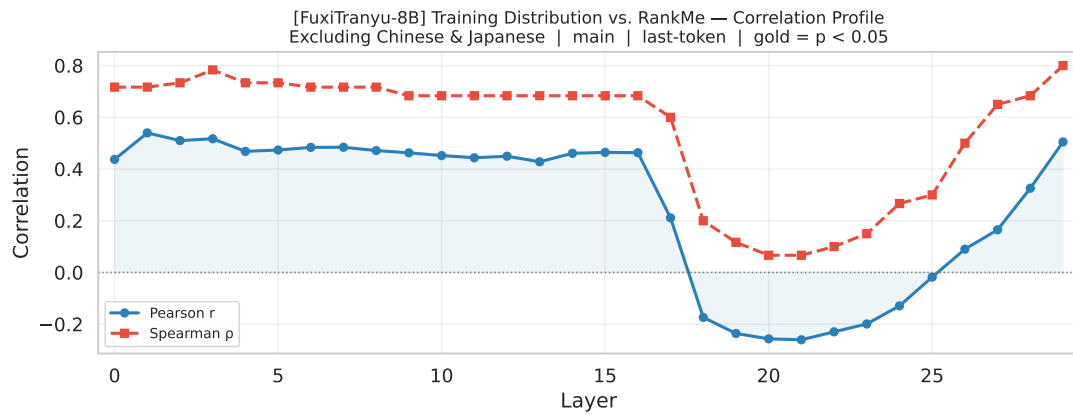


Figure 22: Left: vocabulary coverage per language, meaning the fraction of the vocabulary activated by 5,000 Wikipedia samples; right: correlation between vocabulary coverage and fertility.

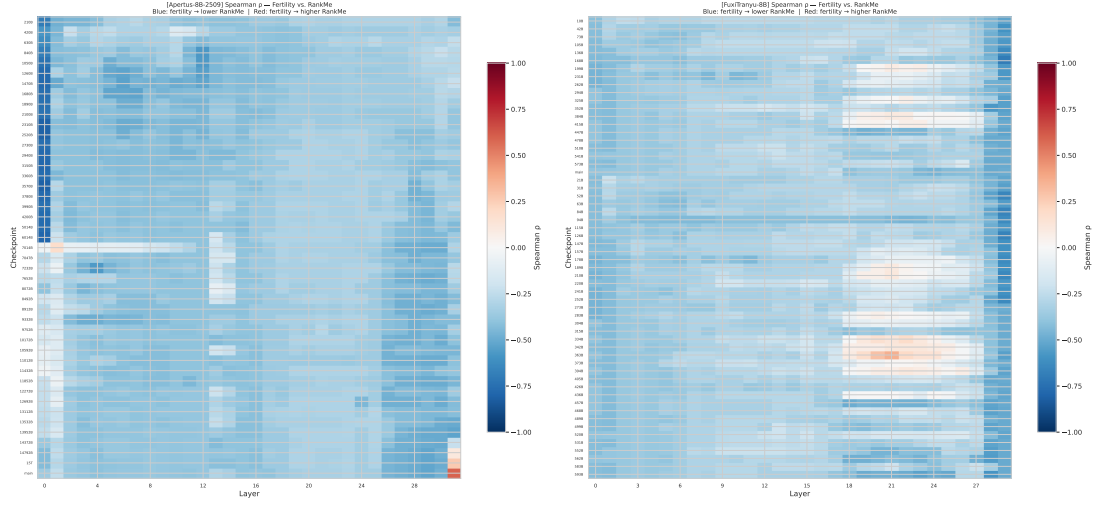


(a) Training distribution vs. RankMe correlation for Fuxi, at last checkpoint and last layer, all languages.

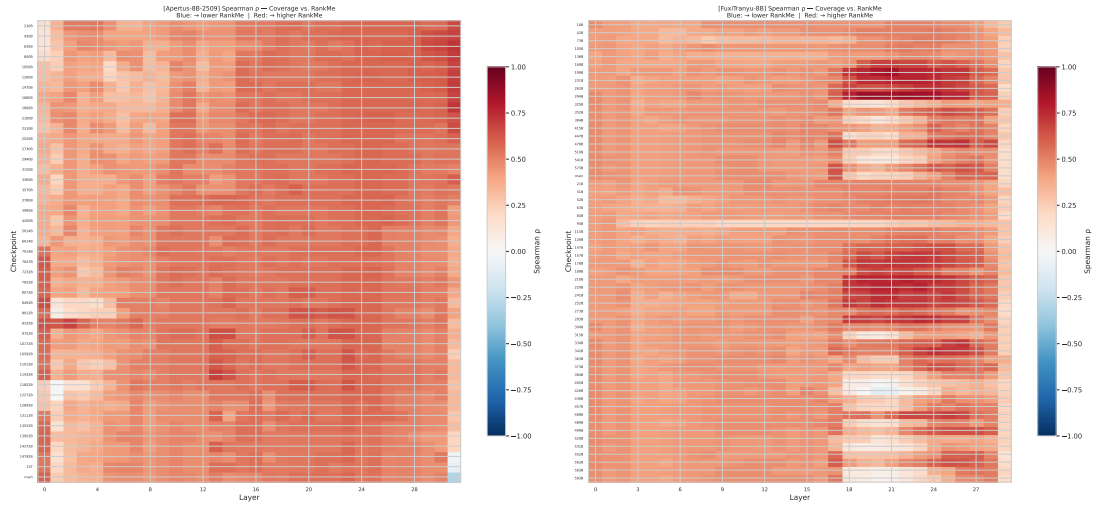


(b) Training distribution vs. RankMe correlation for Fuxi, at last checkpoint and last layer, excluding Chinese and Japanese.

Figure 23: Training distribution vs. RankMe correlation for Fuxi, at last checkpoint and last layer. The correlation values oscillate from positive in early layers, negative in central layers, and back to positive in final layers. Overall, none of the correlations is statistically significant.



(a) Fertility–RankMe Spearman ρ heatmap (left: all languages; right: excluding Chinese & Japanese).



(b) Coverage–RankMe Spearman ρ heatmap (left: all languages; right: excluding Chinese & Japanese).

Figure 24: Spearman ρ between tokenization metrics and RankMe at every (checkpoint, layer) pair, for both models. (a) Fertility: blue indicates high fertility $\Rightarrow$ lower RankMe; red indicates the opposite. (b) Vocabulary coverage: red indicates high coverage $\Rightarrow$ higher RankMe; blue indicates the opposite.

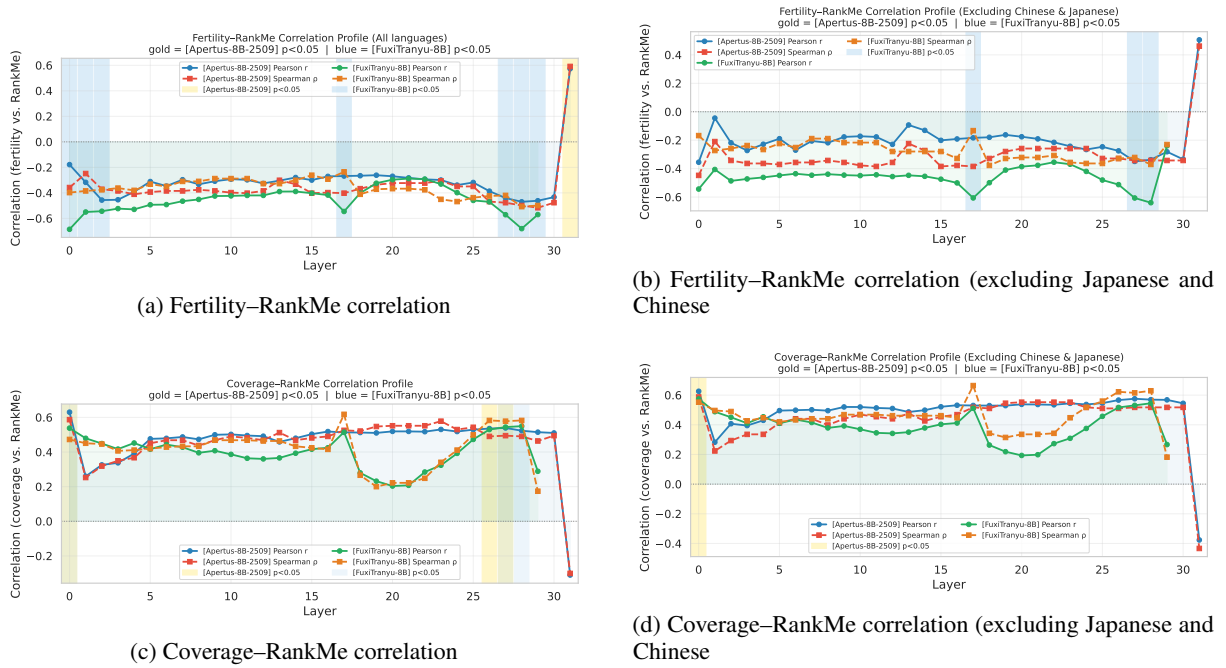


Figure 25: Fertility–RankMe and Coverage–RankMe results for the final checkpoint, last layer, of both models. Results excluding Japanese and Chinese are also present. For Apertus, excluding these languages makes all statistically relevant correlations disappear, posing questions to the actual reliability of the correlations found in the all-languages case.

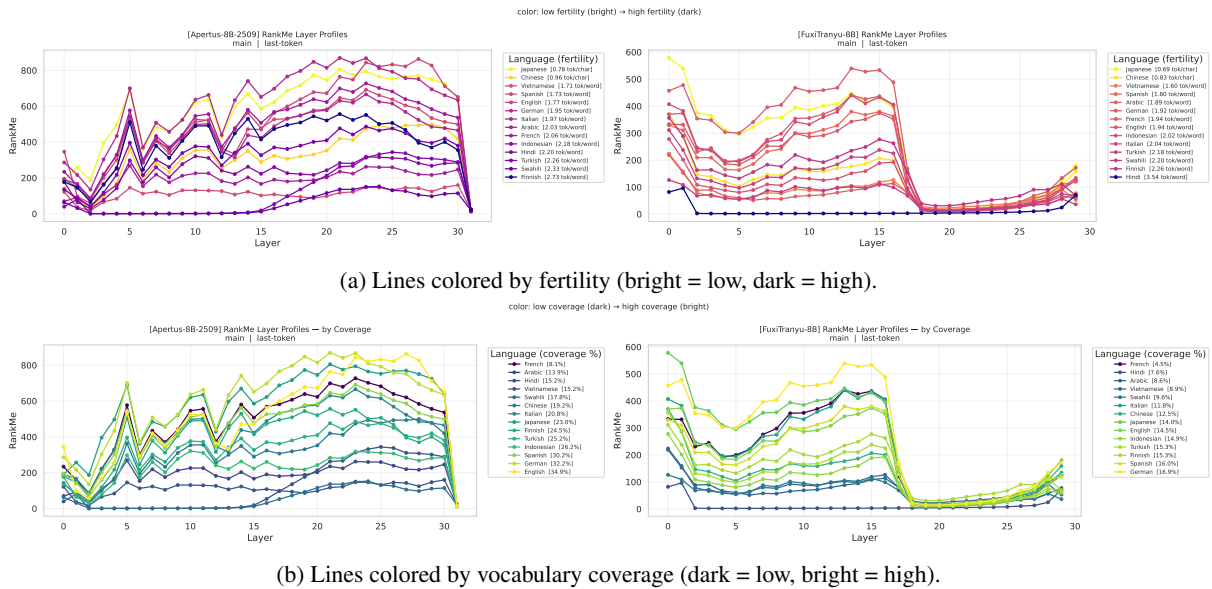


Figure 26: Per-language RankMe trajectories across training checkpoints for both models. In neither plot does the rank ordering of languages in RankMe space mirror their ordering by the coloring variable, indicating that neither fertility nor vocabulary coverage reliably structures representational geometry during training.

D RankMe and Cross-Lingual Transfer

D.1 ECLeKTic

ECLeKTic is a closed-book QA benchmark. Questions are sourced from Wikipedia articles that exist in only one language (so the model likely learned that knowledge monolingually), then translated into 12 languages. We originally planned to use this benchmark to measure cross-lingual transfer, but we found its estimates of the transfer abilities too noisy to be reliable correlation targets, as described below.

Transfer score. The key metric is the transfer score, which measures the fraction of questions answered correctly in the source language that are also answered correctly in the target language. Following the benchmark’s requirements, we evaluate instruction-tuned model variants. Correctness is determined by three LLM judges (gpt-5-mini, meta-llama/Llama-3.3-70B-Instruct, zai-org/GLM-4.7-Flash), each asked whether the model’s response is supported by the article text. A majority vote ($\geq 2/3$ judges) decides the final label.

Reliability of transfer scores. As shown in Table 2, both models answer very few source-language questions correctly, often just one or two. Since the transfer score is a ratio over $2|Q_s^+|$, this yields at most three distinct values per language. With such small sample sizes, the estimates are also highly noisy, making the scores too unreliable to serve as meaningful correlation targets.

Source lang	$ Q_s^+ $ (A)	$ Q_s^+ $ (F)	Unique (A)	Unique (F)
de	2	1	3	1
en	8	3	3	3
es	4	1	3	2
fr	2	1	2	1
hi	6	1	3	2
id	4	1	3	1
it	4	2	3	2
ja	4	2	2	2
zh	6	4	3	2

Table 2: Number of correctly answered source-language questions ($|Q_s^+|$) and number of distinct transfer score values per source language for each model. Column headers are abbreviated for space: A = Apertus, F = FuxiTranyu; “Unique” refers to the number of distinct transfer score values.

D.2 XNLI

D.2.1 Detailed Results

Figure 27 shows the Spearman ρ between RankMe and XNLI accuracy across languages at every training checkpoint when using $k = 8$ examples. Filled markers indicate statistically significant correlations ($p < 0.05$), assessed via permutation testing rather than standard parametric tests, since the same languages appear in multiple pairs and thus violate independence assumptions.

While the aggregated correlation over all checkpoints is about -0.6, we find that individual checkpoints often achieve strong negative correlations between -0.7 and -0.8. Interestingly, predicting XNLI performance $T = 30$ checkpoints ahead achieves some of the strongest correlations. For FuxiTranyu, we do not find any moderate or strong correlations at any checkpoint.

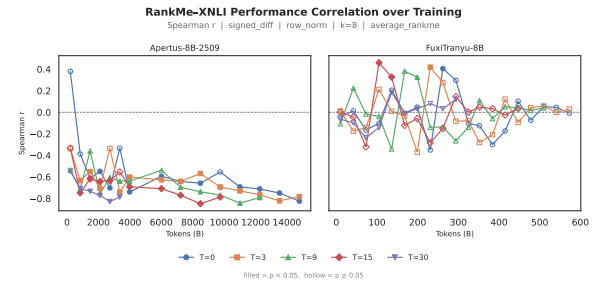


Figure 27: Spearman ρ between the signed RankMe difference and row-normalized XNLI transfer accuracy at every training checkpoint, for $k = 8$ and $T \in \{0, 10, 20, 30\}$ checkpoints ahead. Filled markers indicate $p < 0.05$ (permutation test). Apertus (left) shows strong, consistent negative correlations; FuxiTranyu (right) does not.

D.2.2 Lack of Correlation in FuxiTranyu

We investigate the cause of FuxiTranyu’s flat correlations (Figure 5). We find that accuracy stays near chance ($1/3$) across all 100 language pairs throughout training, so there is no signal to correlate with. Figure 28 indeed shows that accuracy standard deviation never exceeds 0.02, versus up to 0.05 for Apertus.

We also find that the near-chance accuracy is not specific to transfer. Figure 29 shows that same-language accuracy ($\text{src} = \text{tgt}$) is also near chance at the final checkpoint, so FuxiTranyu simply does not learn NLI via ICL at all.

We attribute the issue to training scale: FuxiTranyu was trained on 600B tokens versus Apertus’s 15T, and even Apertus shows near-uniform

accuracy until well past 200B tokens (Figure 28). Meaningful variance across language pairs only emerges after several trillion tokens.

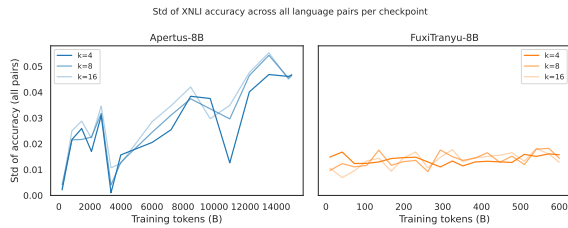


Figure 28: Standard deviation of XNLI accuracy across all language pairs at each training checkpoint, for $k \in \{4, 8, 16\}$. FuxiTranyu (right) stays near zero throughout training, while Apertus (left) develops substantial variance as training progresses.

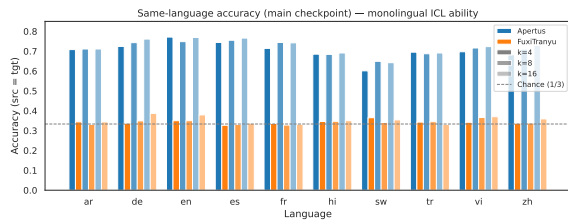


Figure 29: Same-language XNLI accuracy ($\text{src} = \text{tgt}$) at the final checkpoint for both models, across all languages and $k \in \{4, 8, 16\}$. FuxiTranyu’s accuracy is near chance for every language, ruling out cross-lingual transfer as the explanation.

D.2.3 Ablations Studies

We conducted ablation studies to test the robustness of the RankMe–XNLI correlation to various design choices, including correlation coefficient, number of examples, and definition of predictors. We show results for both models for completeness, but we only analyze those for Apertus, where the correlation is stronger and significant.

Correlation coefficient. Our main results use Spearman ρ , which is robust to outliers and monotone non-linearities. Figure 30 shows that switching to Pearson r yields a qualitatively identical pattern, confirming the RankMe–transfer relationship is not an artefact of the rank-based measure.

Choice of predictor. Our main predictor is the signed difference $\text{src} - \text{tgt}$, which is directional. Figure 31 compares it with three alternatives: the normalized asymmetry $(\text{src} - \text{tgt})/(\text{src} + \text{tgt})$, the log ratio $\log(\text{src}/\text{tgt})$, and the unsigned absolute difference $|\text{src} - \text{tgt}|$. The three asymmetric

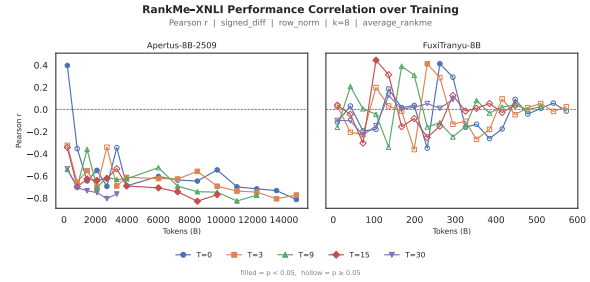


Figure 30: Pearson correlation between RankMe and XNLI accuracy. The pattern mirrors Figure 27, confirming robustness to the choice of correlation coefficient.

predictors yield nearly identical correlations, while the symmetric $|\text{src} - \text{tgt}|$ does not, confirming that direction matters.

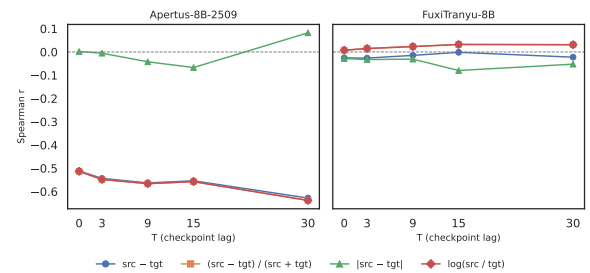


Figure 31: Spearman correlations as a function of checkpoint lag T for four RankMe-based predictors. The three asymmetric predictors ($\text{src} - \text{tgt}$, $(\text{src} - \text{tgt})/(\text{src} + \text{tgt})$, $\log(\text{src}/\text{tgt})$) yield nearly identical results, while the symmetric $|\text{src} - \text{tgt}|$ does not.

Layer and checkpoint aggregation strategy.

Predictors are computed from RankMe values averaged across a range of layers and checkpoints (*average_rankme*). An alternative is to compute predictors independently per layer and checkpoint and then average the resulting scores (*average_predictors*). Figure 32 shows correlations with the *average_predictors* strategy, which are very similar to those with the *average_rankme* strategy shown in Figure 5.

Number of in-context examples (k). We use $k = 8$ ICL examples in our main evaluation. Figure 33 replicates the analysis for $k \in \{4, 8, 16\}$, showing moderate or strong correlations in all cases.

Definition of transfer. Our main transfer score is source-normalized: $\tau_{\text{src}}(s, t) = \text{acc}(s, t)/\text{acc}(s, s)$, measuring how well source language s transfers out relative to its own monolingual performance. An alternative is target normalization: $\tau_{\text{tgt}}(s, t) = \text{acc}(s, t)/\text{acc}(t, t)$,

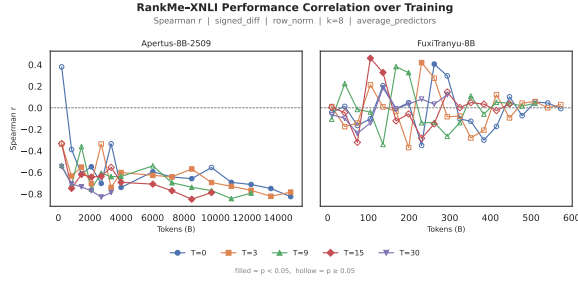


Figure 32: Spearman correlations under the *average_predictors* layer aggregation strategy. Results are nearly identical to the *average_rankme* ones.

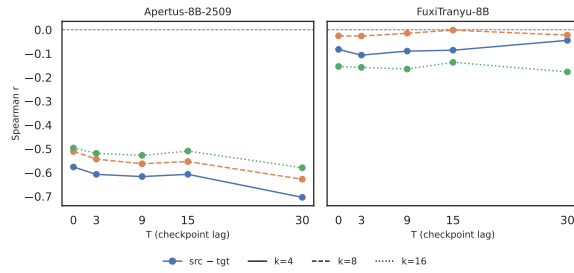


Figure 33: Lag-summary Spearman correlations for $k \in \{4, 8, 16\}$ ICL examples. Results are consistent across shot counts.

which instead measures how much target language t benefits from foreign context relative to its own monolingual performance. Figure 34 shows that correlations under τ_{tgt} are weaker and noisier, with mixed signs and few significant checkpoints. Our predictor $src - tgt$ captures source-side geometry and is naturally aligned with the source-normalized view of transfer as a source property, so weaker results under τ_{tgt} are unsurprising. More broadly, this may suggest that a target language’s ability to receive cross-lingual signal is not primarily governed by the source language’s geometry.

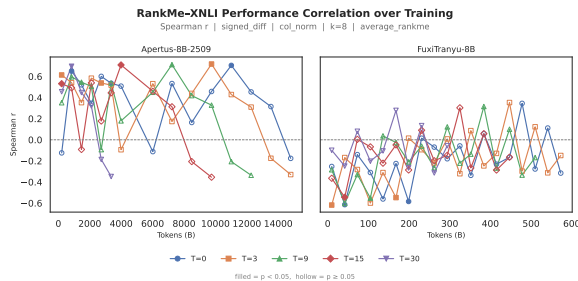


Figure 34: Spearman correlations over training under target-normalized transfer τ_{tgt} . Compared to the source-normalized baseline, correlations are weaker, noisier, and largely insignificant.

D.3 Law Predictors

The strong correlations obtained by averaging RankMe values over multiple checkpoints suggest that the scaling-law parameters estimated in Section A.3 may also serve as useful predictors.

Predictors. The law predictors are static per-language scalars, not tied to any single checkpoint. We use two types of features. The first comes directly from the fitted parametric models: α (the asymptotic floor of the compression phase) and A (the per-language power-law amplitude of the initial compression phase). The second type summarizes the shape of the observed RankMe curve: *drop_to_min* (the magnitude of the initial compression), *recovery* (the magnitude of the subsequent expansion), *drop_minus_recovery* (their difference), and *drop_over_recovery* (their ratio). For each base parameter p and a language pair (s, t) , we report results for two pairwise comparison formulas: the signed difference $p_s - p_t$ and the absolute difference $|p_s - p_t|$.

Correlation setup. We compute Spearman ρ between each law predictor and the source-normalized XNLI transfer score with $k = 8$ ICL examples, pooled across all language pairs. Since the law predictors are static, there is no lag dimension; instead, we report the per-checkpoint correlation over training time, measuring how well the static geometry of the fitted laws predicts transfer at each point in training.

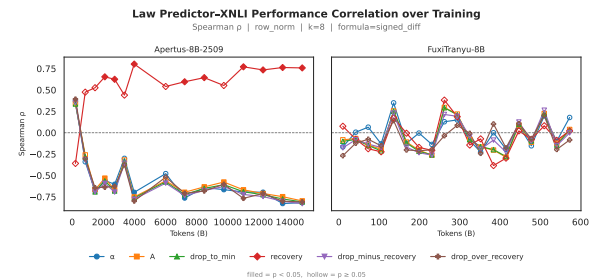


Figure 35: Per-checkpoint Spearman ρ between law-derived predictors (signed-difference formula) and source-normalized XNLI transfer ($k = 8$). Several parameters reach $|\rho| \approx 0.8$ for Apertus; FuxiTranyu shows no significant signal.

Results. Figures 35 and 36 show the per-checkpoint Spearman ρ for the six base parameters, using the signed-difference and absolute-difference comparison formulas respectively. With the signed-difference formula (Figure 35), Apertus

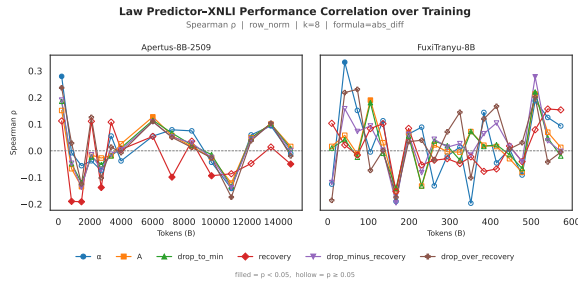


Figure 36: Per-checkpoint Spearman ρ between law-derived predictors (absolute-difference formula) and source-normalized XNLI transfer ($k = 8$). The signed-difference formula (Figure 35) is more informative, as the unsigned formulas discard the direction of the asymmetry.

shows strong correlations for several parameters, reaching $|\rho| \approx 0.8$ at peak checkpoints, matching the magnitude of the best checkpoint-based predictors. Notably, α is a good predictor and requires only early-phase data: when the source language compresses to a higher RankMe floor than the target ($\alpha_s > \alpha_t$), transfer tends to be weaker. The absolute-difference formula (Figure 36) is far less informative ($|\rho| < 0.3$ everywhere), confirming that the direction of the asymmetry is what matters. For FuxiTranyu, all law predictors remain weak ($|\rho| < 0.4$) and mostly insignificant across checkpoints, consistent with the failure of the checkpoint-based predictors on that model.

E RankMe Predicts Downstream Monolingual Performance

This section provides supplementary details for Section 4.4. It covers layer selection methodology, per-language RankMe–accuracy overlay plots for all benchmarks on both models, extended per-checkpoint correlation analyses, and a replication of the main findings using AlphaReQ as a complementary spectral metric.

E.1 Layer Selection

Rather than using a single fixed layer, we select the most informative layer per model by measuring how strongly RankMe separates languages at each depth. For each layer, we average RankMe per language across the first $N=10$ training checkpoints and compute the standard deviation of those per-language averages across all 14 languages. A high standard deviation indicates that the layer encodes language-specific geometry rather than features shared by all languages — making it the most predictive layer for per-language downstream outcomes. Using only the first $N=10$ checkpoints (approximately 17% of FuxiTranyu training and 22% of Apertus training) simulates the practical scenario where predictions must be made early in training, before the full trajectory is available.

Figure 37 shows the stratification signal across all layers for both models. For FuxiTranyu, the signal peaks at **layer 21** ($\sigma=270.3$), located in the upper-middle of the network. For Apertus, it peaks at **layer 19** ($\sigma=204.6$), also in the middle of the network. These are the layers used for all subsequent correlation and per-checkpoint analyses.

Figure 38 shows the RankMe predictor–outcome heatmap for FuxiTranyu, complementing Figure 4 (Apertus) in the main body. As discussed in Section 4.4, almost no cells reach significance for FuxiTranyu: m-MMLU and Belebele accuracy stays at random chance throughout training, and XCOPA is structurally underpowered ($n=4$). M-ARC early-training predictors are the only exception, showing Pearson significance despite non-significant Spearman.

E.2 RankMe–Accuracy Overlay Plots

Figures 39–46 show the full per-language RankMe–accuracy overlay plots for all four benchmarks on both models. Apertus results are shown in Figures 39–42; FuxiTranyu results in Figures 43–46. Each panel overlays the RankMe trajectory (left

axis, blue) and downstream accuracy (right axis, red dashed) across training checkpoints. A dotted horizontal line marks the random-chance baseline (0.25 for 4-way tasks; 0.50 for XCOPA, 2-way) and a purple dashed vertical line marks grokking onset where detected.

E.3 Per-Checkpoint Correlation: Does Geometry Track the Language Performance Hierarchy?

Figure 47 shows the per-checkpoint Spearman ρ between RankMe and task accuracy across languages for both models. For Apertus, the RankMe language ranking tracks the accuracy ranking throughout most of training on m-MMLU, Belebele, and M-ARC; XCOPA shows no significant checkpoints. For FuxiTranyu, significant checkpoints are rare across all benchmarks: m-MMLU, XCOPA, and Belebele produce essentially no signal because accuracy stays at or near random chance. M-ARC is the sole exception (6/57 significant checkpoints, max $\rho=0.817$), consistent with the static predictor results and the cross-model M-ARC replication reported in the main body.

E.4 AlphaReQ: Consistent Results with a Complementary Spectral Metric

AlphaReQ (α) estimates the power-law exponent of the singular value spectrum of the activation matrix. Lower α indicates more structured, organised representations, making it the directional mirror of RankMe: where RankMe shows a positive correlation with performance, AlphaReQ shows a negative one. We use the same selected layers as in the RankMe analysis — layer 21 for FuxiTranyu and layer 19 for Apertus — to ensure direct comparability between the two metrics.

Figure 48 shows the AlphaReQ predictor–outcome heatmap for both models side by side. The pattern mirrors Figure 4: significant peak accuracy cells are negative and shown in blue (lower $\alpha \Leftrightarrow$ higher accuracy), while significant grokking onset cells are positive and shown in red (lower $\alpha \Leftrightarrow$ earlier grokking). The same benchmarks and models reach significance as in the RankMe analysis. For **Apertus**, all 7 AlphaReQ predictors are significant for peak accuracy on m-MMLU, Belebele, and M-ARC (Sp $\rho=-0.82$ – -0.87 , $p\leq 0.008$). First-checkpoint AlphaReQ also strongly predicts grokking onset on all three benchmarks (Sp $\rho=+0.85$, $+0.95$, $+0.83$ respectively), confirming that lower α at the start of training im-

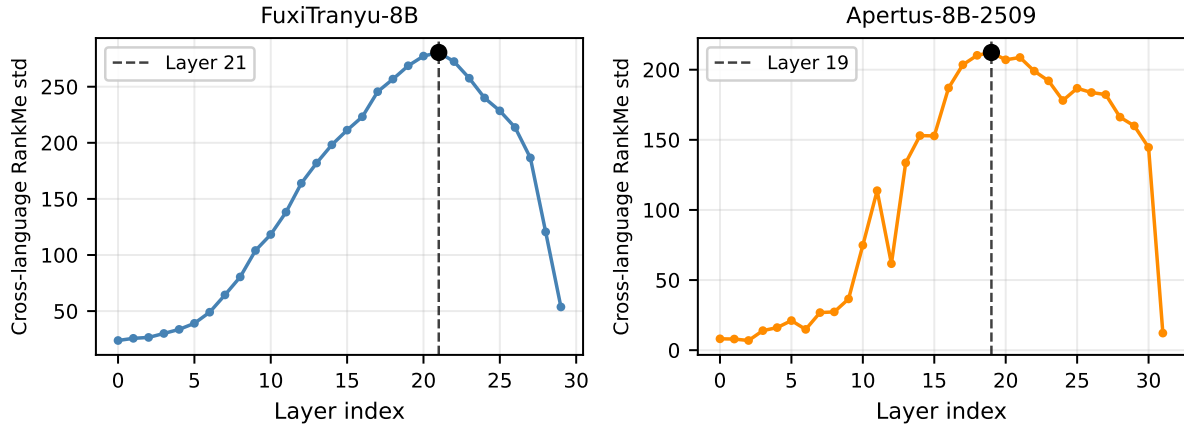


Figure 37: Cross-language RankMe standard deviation per layer, averaged over the first 10 training checkpoints, for FuxiTranyu (left) and Apertus (right). The dashed vertical line marks the selected layer: layer 21 for FuxiTranyu and layer 19 for Apertus, both peaks of the stratification signal.

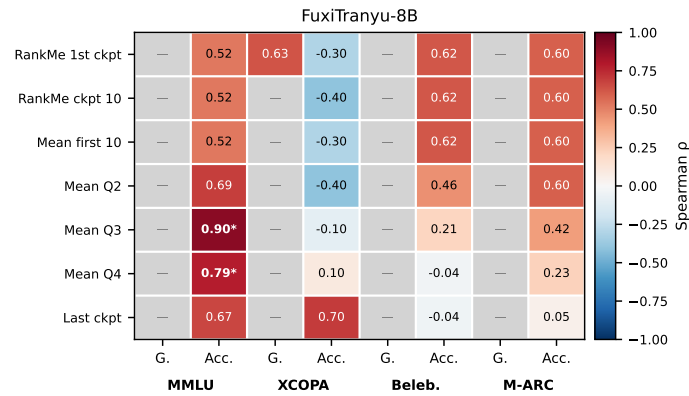


Figure 38: Spearman ρ heatmap for FuxiTranyu (same layout as Figure 4). Gray cells in the grokking onset (G.) columns: grokking onset uses only the first-checkpoint predictor to avoid post-grokking leakage, so the remaining 6 predictors are not applicable. Almost no significant predictors: m-MMLU and Belebele accuracy stays near random chance, and XCOPA grokking onset has only $n=4$ languages (underpowered). M-ARC peak accuracy shows Pearson significance for the four early-training predictors (RankMe at first checkpoint: $r=+0.82$, $p=0.007$; RankMe at checkpoint 10: $r=+0.79$, $p=0.011$; mean RankMe over first 10 checkpoints: $r=+0.78$, $p=0.013$; mean RankMe over Q2, 25–50% of training: $r=+0.76$, $p=0.018$), not captured by the Spearman ρ shown here.

plies earlier grokking. XCOPA shows no signal for Apertus, consistent with the tightly clustered outcomes (0.70–0.78) described in the main body. For **FuxiTranyu**, M-ARC remains the sole cross-model replication (Sp $\rho=-0.667$, $p=0.050$; all 7 predictors significant by Pearson, Pe $r\approx-0.76$ – -0.79), mirroring the positive RankMe–M-ARC relationship.

Figure 49 shows the per-checkpoint Spearman ρ between AlphaReQ and task accuracy for Apertus (left) and FuxiTranyu (right). The results are the direct negative mirror of the RankMe per-checkpoint analysis (Figure 47): for Apertus, 42/44 checkpoints are significant for m-MMLU and Belebele and 44/44 for M-ARC, all with negative ρ ; XCOPA

shows no reliable signal for either model (Apertus: 0/44; FuxiTranyu: 2/57). For FuxiTranyu, M-ARC achieves 35/57 significant checkpoints, confirming the cross-model replication.

Apertus-8B-2509 — M-MMLU: RankMe vs. accuracy

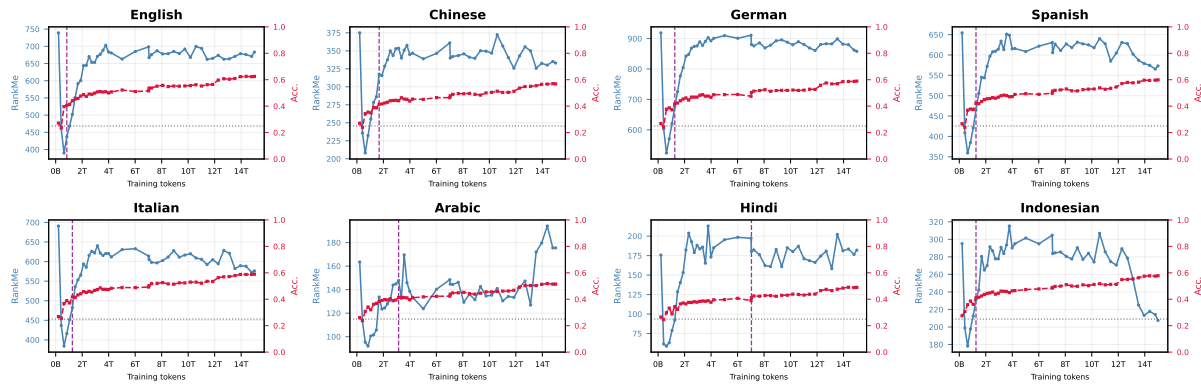


Figure 39: RankMe vs. m-MMLU accuracy for all 8 languages on Apertus. Purple dashed lines mark grokking onset. All 8 languages grokk (840B–7 047B tokens).

Apertus-8B-2509 — XCOPA: RankMe vs. accuracy

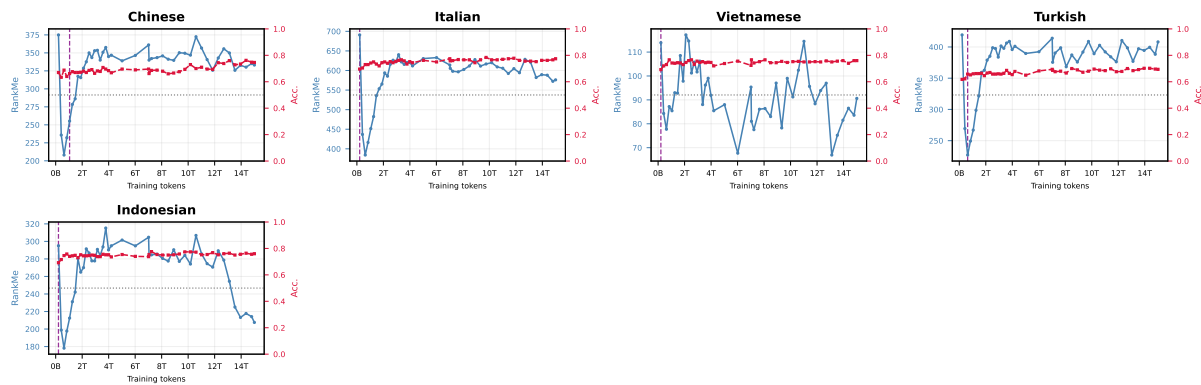


Figure 40: RankMe vs. XCOPA accuracy for all 5 languages on Apertus. All 5 grokk (210B–1 050B tokens); grokking occurs early and outcomes cluster tightly (0.70–0.78).

Apertus-8B-2509 — BELEBELE: RankMe vs. accuracy

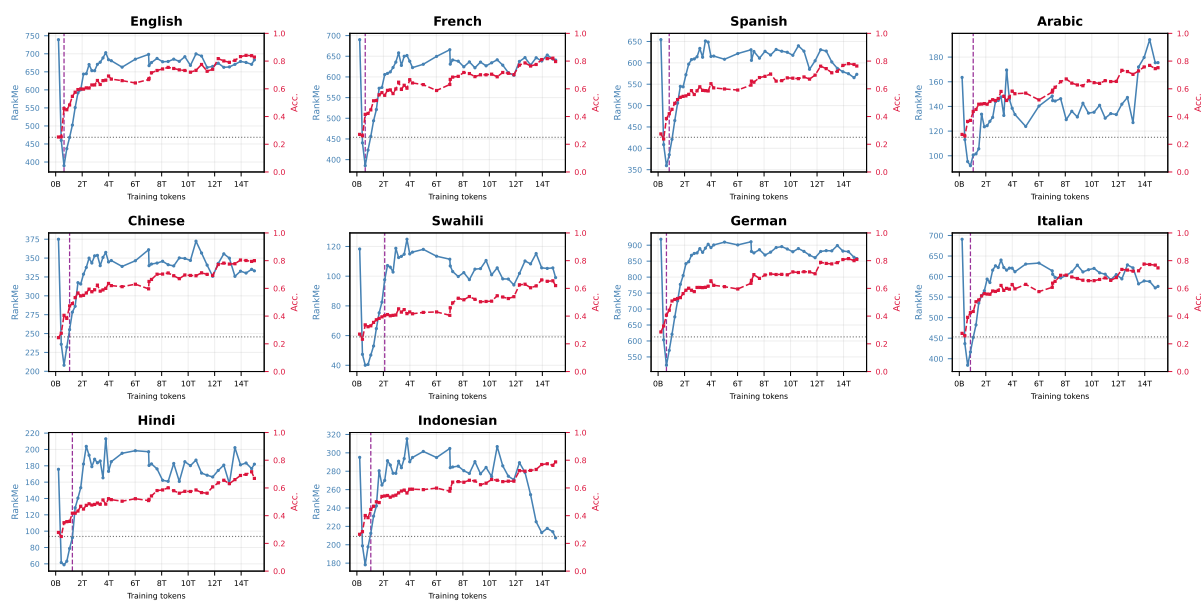


Figure 41: RankMe vs. Belebele accuracy for all 10 languages on Apertus. All 10 grokk (630B–2 100B tokens).

Apertus-8B-2509 — M-ARC: RankMe vs. accuracy

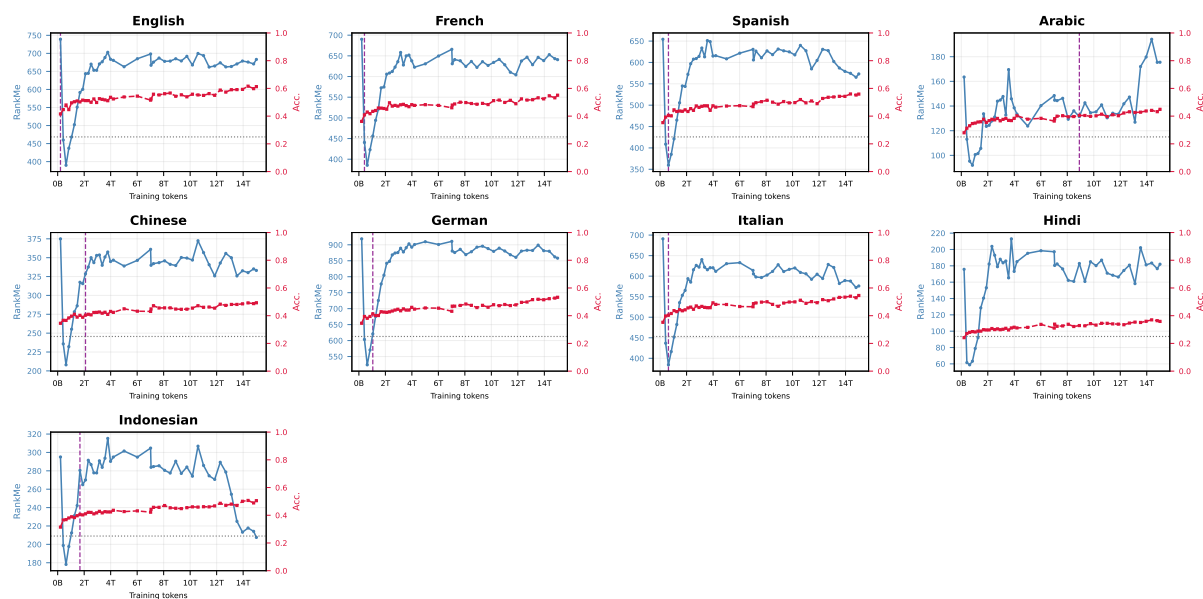


Figure 42: RankMe vs. M-ARC accuracy for all 9 languages on Apertus. 8 of 9 languages grokk (Hindi does not); Arabic grokks late (8 912B tokens).

FuxiTranyu-8B — M-MMLU: RankMe vs. accuracy

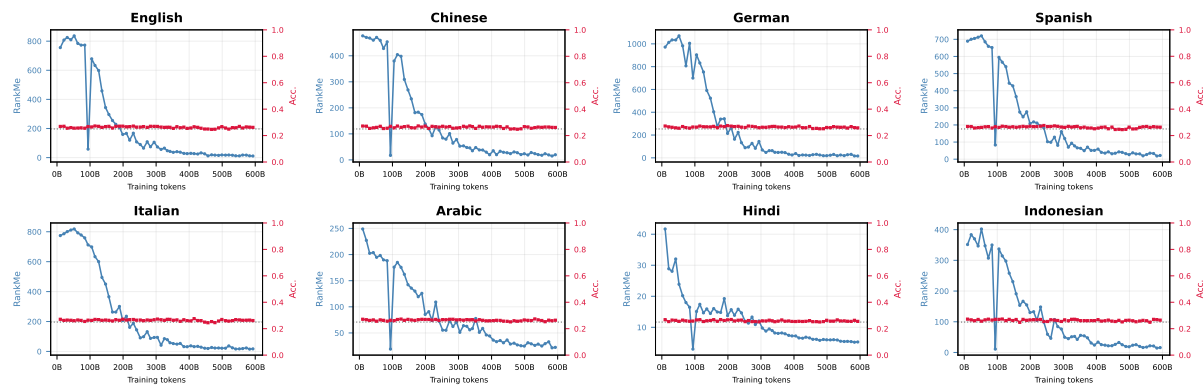


Figure 43: RankMe vs. m-MMLU accuracy for all 8 languages on FuxiTranyu. No language grokks; accuracy stays near random chance (0.25) throughout training.

FuxiTranyu-8B — XCOPA: RankMe vs. accuracy

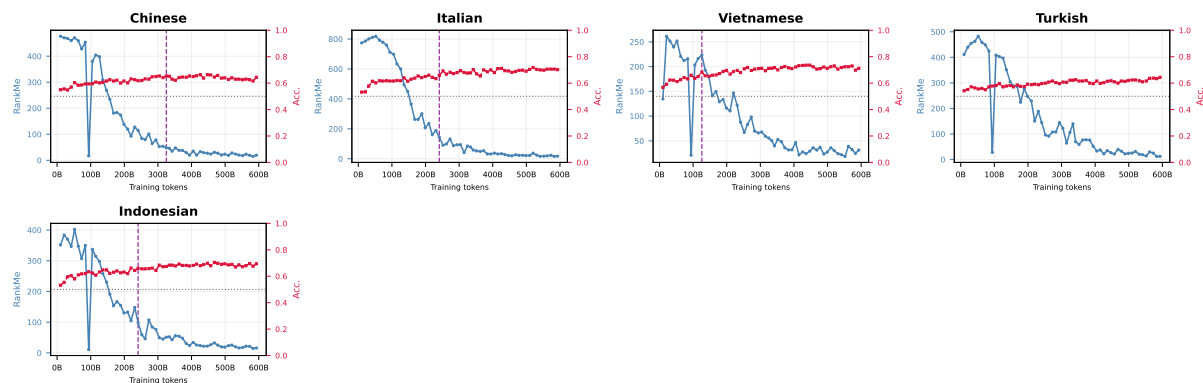


Figure 44: RankMe vs. XCOPA accuracy for all 5 languages on FuxiTranyu. 4 of 5 grokk (Turkish does not); Vietnamese grokks earliest (126B tokens).

FuxiTranyu-8B — BELEBELE: RankMe vs. accuracy

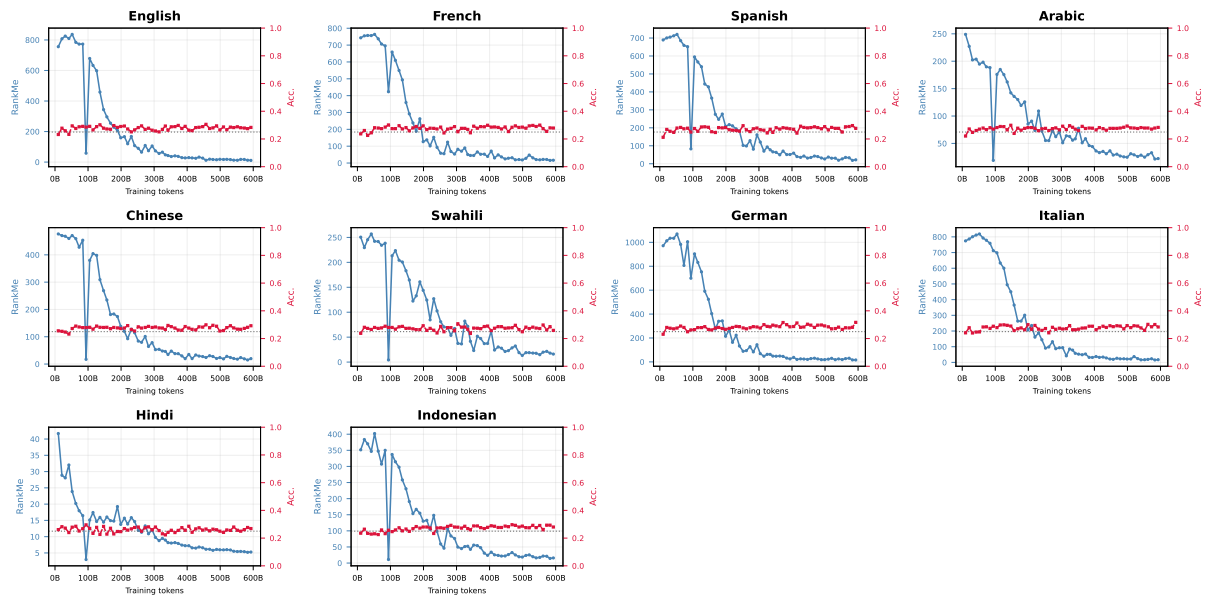


Figure 45: RankMe vs. Belebele accuracy for all 10 languages on FuxiTranyu. No language grokks; accuracy stays near random chance throughout training.

FuxiTranyu-8B — M-ARC: RankMe vs. accuracy

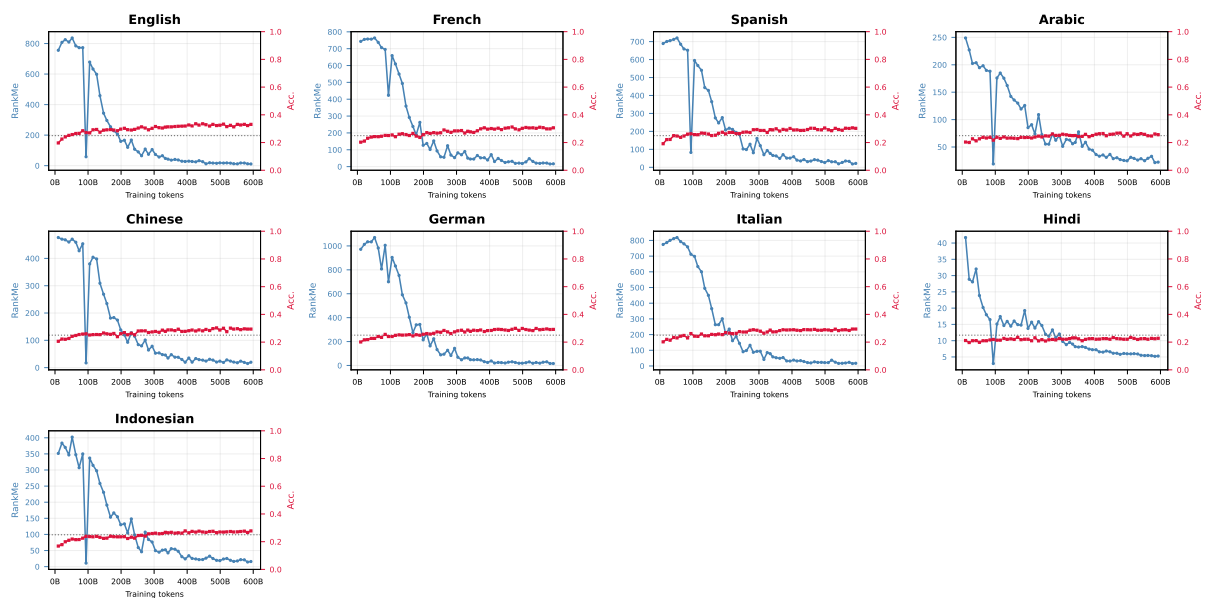


Figure 46: RankMe vs. M-ARC accuracy for all 9 languages on FuxiTranyu. No language grokks; English achieves the highest peak accuracy (0.335).

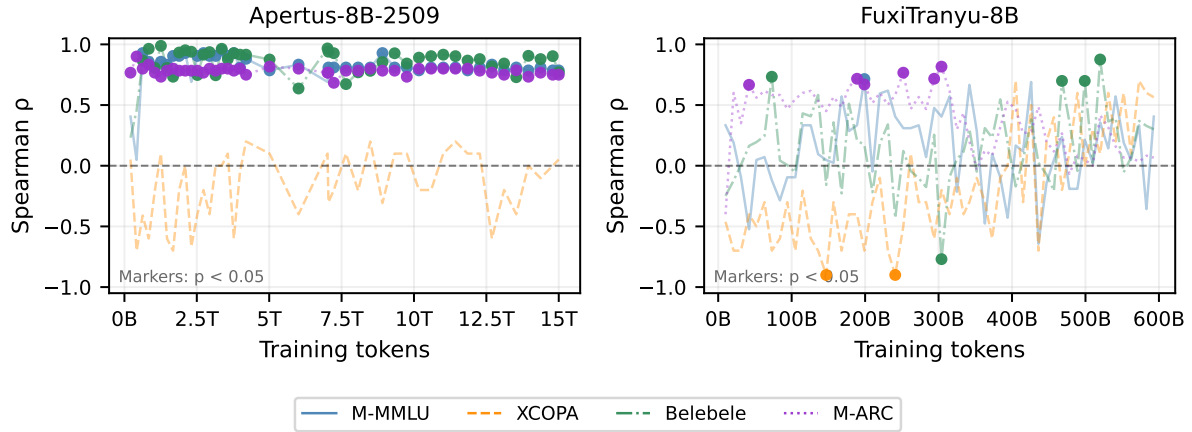


Figure 47: Per-checkpoint Spearman ρ between RankMe and task accuracy across languages on Apertus (left) and FuxiTranyu (right). Markers: $p < 0.05$; line only: $p \geq 0.05$. For FuxiTranyu, significant checkpoints are rare for all benchmarks except M-ARC (6/57).

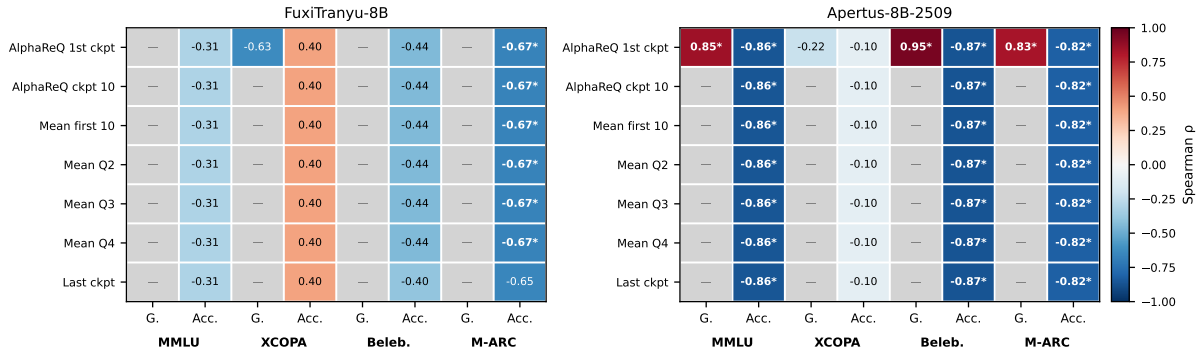


Figure 48: Spearman ρ between AlphaReQ predictors (rows) and downstream outcomes (columns) for FuxiTranyu (left) and Apertus (right). Mirror of Figure 4: significant peak accuracy correlations are negative (blue; lower $\alpha \Leftrightarrow$ higher accuracy), while significant grokking onset correlations are positive (red; lower $\alpha \Leftrightarrow$ earlier grokking). Bold *: statistically significant ($p < 0.05$). Same abbreviations as Figure 4.

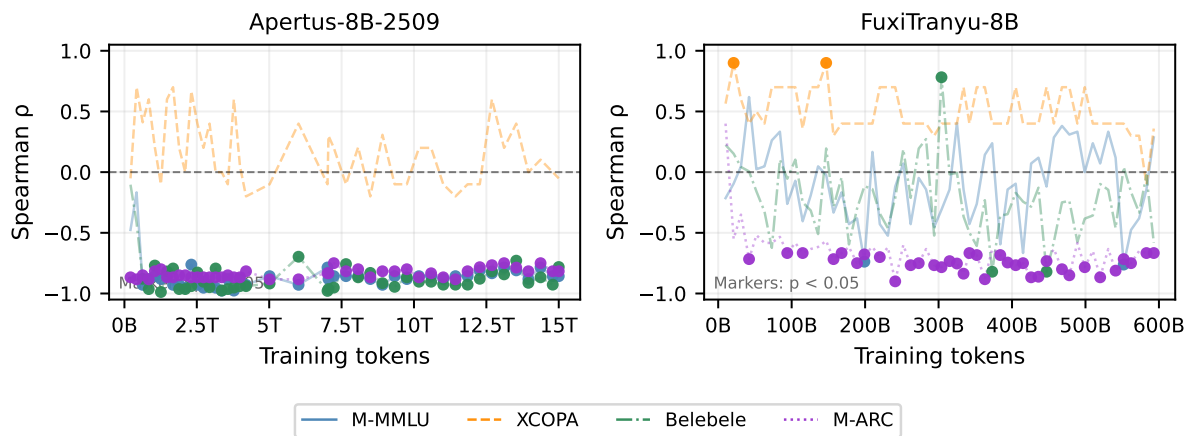


Figure 49: Per-checkpoint Spearman ρ between AlphaReQ and task accuracy across languages on Apertus (left) and FuxiTranyu (right). For Apertus, all significant correlations are negative (lower $\alpha \Leftrightarrow$ better performance), mirroring Figure 47. Markers: $p < 0.05$; line only: $p \geq 0.05$.